DisplayID v2.0 Errata E9
Published 10/7/21

This errata contains all SCRs published through 10/7/21.

The following SCRs are included in DisplayID_v2.0_E9:

- Timing Formula Field Clarification in Type IX Timing Descriptor
- Missing Data Block Payload Length in Table B-1
- Appendix A Detailed Timing Corrections
- Appendix A ColorCorrections
- Support DisplayID version 2 section as a base EDID extension
- Appendix A Color Signaling Capabilities
- Color Depth Mode Support and YCbCr 420 clarification
- DSC pass-through timing support
- DisplayIDv2.0_Dynamic Video Timing Range Limits Data Block_R1
- DisplayID v2.1 Updated type VII timing data block to add future extensibility and accommodate CTA request to explicitly indicate YCC 420 support in each timing descriptor
- DisplayID v2.1 Created a new type X timing data block to support CVT1.3 and support refresh rate field up to 1024 Hz
- DisplayID v2.0 Dark Theme preference with exposure of optional Display Device Theme Preference information
- DisplayID v2.0 Updated Type VIII Timing data block to accommodate CTA request to explicitly indicate YCC 420 support

- DisplayID v2.0 Created a new type X timing data block to support CVT 1.3 RB v3 and support refresh rate field up to 1024 Hz.

- DisplayID v2.0 Mandatory DisplayID Data Blocks and Structure Coexistence

- DisplayID v2.0 new type X timing data block to support CVT 1.3 RB v3 and support refresh rate field up to 1024 Hz Update.

VESA STANDARDS CHANGE REQUEST FORM

To be filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	Timing Formula Field Clarification in Type IX Timing Descriptor
AFFECTED DOCUMENT:	DisplayID v2.0
REVISION CATEGORY:	Category 1
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Bob Ridenour, Apple Inc.

SCR REVISION HISTORY	
(DATE)	**(CHANGE)**
12/12/2017	**Initial Submission of SCR**

(add more rows as needed)

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DATE SCR ADOPTED or REJECTED or DISPOSITIONED	02/02/2018 Adopted

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Summary of the Proposed Change(s)

This change makes the intended references to the CVT specification in the Type IX Timing Descriptor more explicit (including section numbers), without changing the intent.

IPR (Intellectual Property Rights) declaration, if any

None

Benefits as a Result of the Changes

References to timing formulas in the CVT specification will be clear to implementers.

Assessment of the Impact

No expected impact. This change is intended to be purely editorial to clarify references to the timing formulas in the CVT v1.2 specification.

Analysis of the Device Hardware Implications

No known impact to HW implementations.

Analysis of the Device Software Implications

No known impact to SW implementations, unless the intent of the enumerated timing formula references was misunderstood. Any possible impact would be limited to associating each timing formula with a different enumerated value.

Analysis of the Compliance Test & Interop Implications

None.

New Referenced Documents Resulting from Change

None.

Attachments

None.

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Proposed Document Change(s) or Addition(s)

Table 4-21: Type IX Formula-based Timing Descriptor Format

Byte #	Bit #	Description/Format	Flag	Tag
0	**Timing Options**			
	2:0	**Timing Formula/Algorithm** 000b = ~~*CVT Standard v1.2* and higher's standard blanking.~~ CVT Formula - Standard Blanking Timing (*CVT v1.2, section 3.4.1*). 001b = ~~*CVT Standard v1.1* and higher's reduced blanking.~~ CVT Formula - Reduced Blanking Timing Version 1 (*CVT v1.2, section 3.4.2*). 010b = ~~*CVT Standard v1.2* and higher's reduced blanking.~~ CVT Formula - Reduced Blanking Timing Version 2 (*CVT v1.2, section 3.4.3*). All other values are RESERVED.		✔

Background Information

It appears that the version number of the reduced blanking formula referenced in the CVT standard got mixed up with the version number of the CVT standard itself. The previous version of the DisplayID specification referred to all three of these formulas, so implementers should be familiar with them, and this change simply makes the expected references to the CVT v1.2 specification very explicit.

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VESA STANDARDS CHANGE REQUEST FORM

To be filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	Missing Data Block Payload Length in Table B-1
AFFECTED DOCUMENT:	DisplayID v2.0
REVISION CATEGORY:	Category 1
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Bob Ridenour, Apple Inc.

SCR REVISION HISTORY	
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04/10/2018	**Initial Submission of SCR**

(add more rows as needed)

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VESA SCR NUMBER:	(To be assigned by VESA office)
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DATE SCR ADOPTED or REJECTED or DISPOSITIONED	05/05/2018 Adopted

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Summary of the Proposed Change(s)

This change inserts a missing header byte in the description of the VESA Organization Vendor-specific Data Block in Table B-1 of Appendix B.

IPR (Intellectual Property Rights) declaration, if any

None

Benefits as a Result of the Changes

The definition of the VESA Organization Vendor-specific Data Block in Table B-1 will include the mandatory Data Block header bytes, avoiding implementer confusion.

Assessment of the Impact

No expected impact. This change is intended to be purely editorial as the missing byte is clearly defined in section 4.9 where the required structure of the Vendor-specific Data Block is specified.

Analysis of the Device Hardware Implications

No known impact to HW implementations.

Analysis of the Device Software Implications

No known impact to SW implementations.

Analysis of the Compliance Test & Interop Implications

None.

New Referenced Documents Resulting from Change

None.

Attachments

None.

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Proposed Document Change(s) or Addition(s)

Table B-1: VESA Organization Vendor-specific Data Block

Offset	Bit #	Description/Format	Tag
00h	7:0	**Vendor-specific Data Block** Value is 7Eh.	✔
01h	**Block Revision and Other Data**		
	2:0	**Block Revision** Revision values range from 0 through 7. 000b = Revision 0 (default).	
	7:3	**RESERVED** Cleared to all 0s.	
02h	7:0	**Number of Payload Bytes in Block** Number of bytes range from 3 through 248. 05h = Payload length	
03h	7:0	**IEEE OUI First Byte** Values range from 0 through 255. 3Ah = First VESA IEEE OUI byte.	
04h	7:0	**IEEE OUI Second Byte** Values range from 0 through 255. 02h = First VESA IEEE OUI byte.	
05h	7:0	**IEEE OUI Third Byte** Values range from 0 through 255. 92h = First VESA IEEE OUI byte.	
06h	**VESA Data Structure Type**		
	2:0	**Data Structure Type** 000b = Embedded DisplayPort (eDP). 001b = External DisplayPort. All other values are RESERVED for future use.	
	6:3	**RESERVED** Cleared to all 0s.	

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	7	**Default Color Space and EOTF Handling** 0 = MISC signaling of “RGB unspecified color space” by a Source device is interpreted by a Sink device as sRGB color space and EOTF. 1 = MISC signaling of “RGB unspecified color space” by a Source device is interpreted by a Sink device as “native” color space and EOTF as specified in the Display Parameters data block.	
07h	3:0	**Number of Pixels in Horizontal Pixel Count Overlapping an Adjacent Panel Segment** Values range from 0 through 8. Value range of 9 through 15 is RESERVED.	
	4	**RESERVED** Cleared to 0.	
	6:5	**Multi-SST Operation** 00b = Not supported (Conventional Single-Stream Transport). 01b = Two streams (number of links shall be two or four). 10b = Four streams (number of links shall be four). 11b = RESERVED for future use.	
	7	**RESERVED** Cleared to 0.	

Background Information

It appears that during the final editing process, the required payload length byte in the Vendor-specific Data Block was inadvertently removed and the bytes at offsets 03h through 06h were moved up. The payload length byte is a required element of the Data Block, so it must be included in order for this VESA Organization Vendor-specific Data Block to be valid.

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VESA STANDARDS CHANGE REQUEST FORM

To be filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	Appendix A Detailed Timing Corrections
AFFECTED DOCUMENT:	DisplayID v2.0
REVISION CATEGORY:	Category 1
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Bob Ridenour, Apple Inc.

SCR REVISION HISTORY	
(DATE)	**(CHANGE)**
04/10/2018	**Initial Submission of SCR**

(add more rows as needed)

To be filled in by VESA Office:

VESA SCR NUMBER:	(To be assigned by VESA office)
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DATE SCR ADOPTED or REJECTED or DISPOSITIONED	06/15/2018 - Adopted

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Summary of the Proposed Change(s)

This change corrects some minor errors in the Type VII detailed timing descriptors that are included in the example DisplayID structure in Appendix A. It also adds the previously unspecified section checksum to the example DisplayID structure in Appendix A.

IPR (Intellectual Property Rights) declaration, if any

None

Benefits as a Result of the Changes

The timings defined in the example in Appendix A will be consistent with the definition of Type VII detailed timings in the DisplayID specification, and CVT Reduced Blanking version 2 in the CVT specification.

Assessment of the Impact

No expected impact. This change is intended to be purely editorial as it corrects some values in an example structure only. The normative parts of the DisplayID and CVT specs that are relevant to the example are not affected.

Analysis of the Device Hardware Implications

No known impact to HW implementations.

Analysis of the Device Software Implications

No known impact to SW implementations.

Analysis of the Compliance Test & Interop Implications

None.

New Referenced Documents Resulting from Change

None.

Attachments

None.

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Proposed Document Change(s) or Addition(s)

4Bh	0x22	**Data Block Name**	Type VII Detailed Timing Data Block.
4Ch	0x00	**Block Revision and Other Data**	Bits 7:3 = RESERVED = 00000b. Bits 2:0 = Block Revision = 0.
4Dh	0x3C	**Number of Payload Bytes in Block**	Payload Length = 60 bytes.
4Eh 4Fh 50h 51h 52h 53h 54h 55h 56h 57h 58h 59h 5Ah 5Bh 5Ch 5Dh 5Eh 5Fh 60h 61h	0xC~~8~~7 0x7E 0x08 0x88 0xFF 0x0F 0x4F 0x00 0x07 0x80 0x1F 0x00 0x6F 0x08 0x3D 0x00 0x~~00~~2F 0x00 0x07 0x00	**Timing 1**	Pixel Clock = 556.744MP/s. Bit 7 = 1 (Preferred Detailed timing). Bits 6:5 = 00b (Timing always mono – no stereo). Bit 4 = 0 (Progressive scan frame). Bits 3:0 = 1000b (Aspect Ratio calculated from pixel dimensions). Horizontal Active Image Pixels = 4096. Horizontal Blank Pixels = 80. Horizontal Sync Offset (front porch) = 8. Horizontal Sync Polarity = 1 (positive). Horizontal Sync Width = 32. Vertical Active Image Lines = 2160. Vertical Blank Lines = 62. Vertical Sync Offset (front porch) = ~~1~~48. Vertical Sync Polarity = 0 (negative). Vertical Sync Width = 8.

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62h 63h 64h 65h 66h 67h 68h 69h 6Ah 6Bh 6Ch 6Dh 6Eh 6Fh 70h 71h 72h 73h 74h 75h	0x5D 0x94 0x03 0x08 0xFF 0x09 0x4F 0x00 0x07 0x80 0x1F 0x00 0x9F 0x05 0x28 0x00 0x~~00~~1A 0x00 0x07 0x00	**Timing 2**	Pixel Clock = 234.590MP/s. Bit 7 = 0 (Not a preferred detailed timing). Bits 6:5 = 00b (Timing always mono – no stereo). Bit 4 = 0 (Progressive scan frame). Bits 3:0 = 1000b (Aspect Ratio calculated from pixel dimensions). Horizontal Active Image Pixels = 2560. Horizontal Blank Pixels = 80. Horizontal Sync Offset (front porch) = 8. Horizontal Sync Polarity = 1 (positive). Horizontal Sync Width = 32. Vertical Active Image Lines = 1440. Vertical Blank Lines = 41. Vertical Sync Offset (front porch) = ~~1~~27. Vertical Sync Polarity = 0 (negative). Vertical Sync Width = 8.
76h 77h 78h 79h 7Ah 7Bh 7Ch 7Dh 7Eh 7Fh 80h 81h 82h 83h 84h 85h 86h 87h 88h 89h	0xC7 0x08 0x02 0x08 0x7F 0x07 0x4F 0x00 0x07 0x80 0x1F 0x00 0x37 0x04 0x1E 0x00 0x~~00~~10 0x00 0x07 0x00	**Timing 3**	Pixel Clock = 133.320MP/s. Bit 7 = 0 (Not a preferred detailed timing) Bits 6:5 = 00b (Timing always mono – no stereo) Bit 4 = 0 (Progressive scan frame) Bits 3:0 = 1000b (Aspect Ratio calculated from pixel dimensions). Horizontal Active Image Pixels = 1920. Horizontal Blank Pixels = 80. Horizontal Sync Offset (front porch) = 8. Horizontal Sync Polarity = 1 (positive). Horizontal Sync Width = 32. Vertical Active Image Lines = 1080. Vertical Blank Lines = 31. Vertical Sync Offset (front porch) = ~~1~~17. Vertical Sync Polarity = 0 (negative). Vertical Sync Width = 8.

8Ah	0x66	**Checksum Section**	

Background Information

The pixel rate specified in the raw data of the first Type VII timing in the example structure is off by one. The description of the pixel rate, however, is correct.

All three of the Type VII timings in the example structure are intended to be compliant with the CVT Reduced Blanking version 2 standard. However, when the example was generated, the Vertical Sync Offset was generated incorrectly. This field needs to be corrected for all three timings in order to match the correct CVT generated timing.

The section checksum was previously not specified. It is now included for the example structure.

- End of Document -

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VESA STANDARDS CHANGE REQUEST FORM

To be filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	Appendix A Color Corrections
AFFECTED DOCUMENT:	DisplayID v2.0
REVISION CATEGORY:	Category 1
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Bob Ridenour, Apple Inc.

SCR REVISION HISTORY	
(DATE)	**(CHANGE)**
10/09/2018	**Initial Submission of SCR**

(add more rows as needed)

To be filled in by VESA Office:

VESA SCR NUMBER:	(To be assigned by VESA office)
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SCR ADOPTED, REJECTED, or otherwise DISPOSITIONED for other action	SCR is (adopted) or (rejected) or (Dispositioned for other action) If rejected, explain reason for acceptation or rejection If dispositioned, explain action or plan for action (such as including in future draft specification revision, or re-visiting at future date, or other)
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Summary of the Proposed Change(s)

This change corrects a minor error in the Display Parameters Data Block that is included in the example DisplayID structure in Appendix A. The raw value of the Native Color Chromaticity (Green) field is corrected to accurately reflect the interpretation.

IPR (Intellectual Property Rights) declaration, if any

None

Benefits as a Result of the Changes

The interpretation of the Chromaticity fields in the example in Appendix A will be consistent with the raw values provided in the example.

Assessment of the Impact

No expected impact. This change is purely editorial as it corrects a value in an example structure only.

Analysis of the Device Hardware Implications

No known impact to HW implementations.

Analysis of the Device Software Implications

No known impact to SW implementations.

Analysis of the Compliance Test & Interop Implications

None.

New Referenced Documents Resulting from Change

None.

Attachments

None.

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Proposed Document Change(s) or Addition(s)

2Eh 2Fh 30h	0x45 0x~~5~~64 0xA6	**Native Color Chromaticity (Green)**	Green X = 0.267, Y = 0.650.

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Background Information

The example DisplayID structure in Appendix A includes a Display Parameters Data Block that includes Native Color Chromaticity values. The native green chromaticity is intended to match the interpretation which is stated in decimal as "Y=0.650". The raw value for this parameter is given in the example as 0xA65. However, the correct raw value converted from 0.650 is 0xA66. This change simply corrects the raw value in the example structure to match the stated interpretation.

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VESA STANDARDS CHANGE REQUEST FORM

To be Filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	Support DisplayID version 2 section as a base EDID extension
AFFECTED DOCUMENT:	DisplayID standard version 2.0
REVISION CATEGORY:	
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Syed Athar Hussain

SCR REVISION HISTORY	
(DATE)	**(CHANGE)**
MM/DD/YYYY	**11/26/18 Version 1**
	12/4/2018 Version 2

(add more rows as needed)

To be Filled in by VESA Office:

VESA SCR NUMBER:	(To be assigned by VESA office)
SCR ENTRY DATE:	11/26/2018

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DATE SCR ADOPTED or REJECTED or DISPOSITIONED	03/17/2019 - Adopted

Summary of the Proposed Change(s)

This SCR adds the necessary framework in the DisplayID version 2.0 specification to enable support of DisplayID version 2 section as a base EDID extension.

IPR (Intellectual Property Rights) declaration, if any

None.

Benefits as a Result of the Changes

DisplayID version 2.0 section can be utilized as base EDID extension.

Assessment of the Impact

SW and/or FW need to be updated to be able to support new information exposed in the Display ID version 2.0 section as EDID based extension.

Analysis of the Device Hardware Implication

No HW impact is expected.

Analysis of the Device Software Implications

Some SW impact is expected to support parsing of the new structure as part of the EDID extension block.

Analysis of the Compliance Test & Interop Implications

Need to be updated to ensure correct DisplayID exposure of the structure by the sink and correct parsing of the source.

New Referenced Documents Resulting from Change

None

Attachments

None

Proposed Document Change(s) or Addition(s)

Table 2-4 shall be updated as described below. Note this change applies to DisplayID extension section and changes described here were editorial in nature.

Table 2-4: DisplayID Extension Section Review

Offset	Bit #	Register
00h	**DisplayID Structure Version 2, Revision 0**	
	3:0	**Revision** 0h = Revision 0.
	7:4	**Version** 0h = RESERVED. 2h = Version 2.
01h	7:0	**Bytes in Section** Section length, excluding the five mandatory bytes. Value is *N* - 5.[a]

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02h	3:0	**Display Product Primary Use Case** Value shall be 0h.
	7:4	RESERVED, cleared to all 0s.
03h	7:0	**Extension Count** Value shall be 00h
04h	7:0	**Data Block** See Section 3.
…		**…**
(N - 1)h[a]		**Checksum** Value ranges from 00h through FFh.

a. N represents the total number of bytes within the DisplayID section.

A new paragraph will be added in section 2 to describe how to set up DisplayID version 2 section as a base EDID extension. The proposed new paragraph is described below.

- Proposed Text Change(s) –

2.1 DisplayID version 2.0 as a base EDID extension

To facilitate usage of DisplayID version 2.0 section as a base EDID extension, a new framework is defined. A Sink with base EDID and utilizing DisplayID version 2.0 section as extension shall populate the listed fields as described below:

- EDID extension block tag shall be set to 70h denoting existence of a DisplayID section
- Following the block tag byte, a regular DisplayID version 2.0 section structure shall be used with a fixed length fill data byte to map to the EDID extension requirement of 128 bytes per extension.
- DisplayID Structure Version/Revision byte shall be set to 20h corresponding to DisplayID version 2 revision 0.
- First DisplayID version 2.0 section as EDID extension shall populate Display Product Primary Use Case byte with a value from 1h-8h based on the intended primary use case of the sink. If the Primary Use Case is not known Sink shall set the value of 2h. Any subsequent DisplayID version 2.0 section EDID extension shall set the Display Product Primary Use Case byte to 0h.
- Extension Count byte shall be set to 00h
- Bytes in Section = Shall be set to 79h (121)
    - This corresponds to 5 additional bytes for the DisplayID section framework + EDID extension block tag for DisplayID and EDID extension block checksum for total of 128 bytes
- EDID extension block checksum shall be calculated using the DisplayID section checksum and extension block tag byte. Since DisplayID section checksum already ensures that all DisplayID section bytes adds up so that modulo 256 of the sums of all byte is 0h, only block tag byte needs to be considered for EDID extension block checksum. Consequently, EDID extension block checksum shall be set to 90h.

The following diagram shows an example of the DisplayID version 2 section as a base EDID extension.

Example: DisplayID Extension to Base EDID – with fill data for fixed length section							
			Field Name	Offset (Dec)	Offset (Hex)	Value (Hex)	Value (Dec)
EDID Extension Format	Display ID 2 EDID Extension	EDID Extension Block Tag	Tag	0	0	70	112
	Display ID Extension Section	Display ID 2 Extension Section Header	Structure Version/Revision	1	0	20	32
			Bytes in Section	2	1	79	121
			Display Product Primary Use Case	3	2	2	2
			Extension Count Reserved (00)	4	3	00	00
		Section Data Block(s) (Example)	Data Block Tag	5	4	ID	ID
			Data Block Revision	6	5	0	0
			Payload Bytes	7	6	6	6
			Payload Bytes #1	8	7	PB1	PB1
			Payload Bytes #2	9	8	PB2	PB2
			Payload Bytes #3	10	9	PB3	PB3
			Payload Bytes #4	11	0A	PB4	PB4
			Payload Bytes #5	12	0B	PB5	PB5
			Payload Bytes #6	13	0C	PB6	PB6
		Fill Data for Fixed Length (Example)	Fill Data #1	14	0D	0	0
			Fill Data #2	15	0E	0	0
			Fill Data #3 ..125	125	7D	0	0
			Section Checksum	126	7E	CHK_DID	CHK_DID
			EDID Extension Block Checksum	127	7F	90	144

Figure X-X: Example of DisplayID version 2 section as a base EDID extension.

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Background Information

Additional background for justifying the SCR can be included here. For example, a presentation prepared for Task Group introduction can be placed in this section.

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VESA STANDARDS CHANGE REQUEST FORM

To be filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	Appendix A Color Signaling Capabilities
AFFECTED DOCUMENT:	DisplayID v2.0
REVISION CATEGORY:	TBD
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Bob Ridenour, Apple Inc.

SCR REVISION HISTORY	
(DATE)	**(CHANGE)**
12/18/2018	**Initial Submission of SCR**
03/05/2019	TGR Comments Addressed

(add more rows as needed)

To be filled in by VESA Office:

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SCR ADOPTED, REJECTED, or otherwise DISPOSITIONED for other action	SCR is (adopted) or (rejected) or (Dispositioned for other action) If rejected, explain reason for acceptation or rejection If dispositioned, explain action or plan for action (such as including in future draft specification revision, or re-visiting at future date, or other)
DATE SCR ADOPTED or REJECTED or DISPOSITIONED	04/22/2019 Adopted

Summary of the Proposed Change(s)

This change adds a VESA Organization Vendor-specific Data Block to the example DisplayID structure in Appendix A in order to clarify how the color space of received pixel data is interpreted by the Sink that is described by the example structure. Additional guidance regarding implied or declared color space signaling is also added.

IPR (Intellectual Property Rights) declaration, if any

None

Benefits as a Result of the Changes

Clarifies how DisplayPort Sources should interpret the capabilities in the example structure, and encourages use of the VESA Organization Vendor-specific Data Block in DisplayID structures in order to remove ambiguity related to use of some color space signaling in DisplayPort.

Assessment of the Impact

No expected impact. This change is editorial in that it modifies the example structure, but not the normative text.

Analysis of the Device Hardware Implications

No known impact to HW implementations.

Analysis of the Device Software Implications

No known impact to SW implementations.

Analysis of the Compliance Test & Interop Implications

None.

New Referenced Documents Resulting from Change

None.

Attachments

None.

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Proposed Document Change(s) or Addition(s)

76h 77h 78h 79h 7Ah 7Bh 7Ch 7Dh 7Eh 7Fh 80h 81h 82h 83h 84h 85h 86h 87h 88h 89h	0xC7 0x08 0x02 0x08 0x7F 0x07 0x4F 0x00 0x07 0x80 0x1F 0x00 0x37 0x04 0x1E 0x00 0x10 0x00 0x07 0x00	**Timing 3**	Pixel Clock = 133.320MP/s. Bit 7 = 0 (Not a preferred detailed timing) Bits 6:5 = 00b (Timing always mono – no stereo) Bit 4 = 0 (Progressive scan frame) Bits 3:0 = 1000b (Aspect Ratio calculated from pixel dimensions). Horizontal Active Image Pixels = 1920. Horizontal Blank Pixels = 80. Horizontal Sync Offset (front porch) = 8. Horizontal Sync Polarity = 1 (positive). Horizontal Sync Width = 32. Vertical Active Image Lines = 1080. Vertical Blank Lines = 31. Vertical Sync Offset (front porch) = 17. Vertical Sync Polarity = 0 (negative). Vertical Sync Width = 8.
8Ah	0x7E	**Data Block Name**	Vendor-specific Data Block
8Bh	0x00	**Block Revision and Other Data**	Bits 7:3 = RESERVED = 00000b. Bits 2:0 = Block Revision 0 = 000b.
8Ch	0x05	**Number of Payload Bytes**	Data block is composed of 5 payload bytes.
8Dh 8Eh 8Fh	0x3A 0x02 0x92	**Manufacturer/Vendor ID**	IEEE OUI = 3A-02-92
90h	0x81	**VESA Data Structure Type**	Bit 7 = Default Color Space and EOTF = 1 (RGB unspecified color space is interpreted by the Sink device as native color space and EOTF as specified in the Display Parameters data block) Bits 6:3 = RESERVED = 000b. Bits 2:0 = Data Structure Type = 001b (External DisplayPort)
91h	0x00	**Multi-SST Operation**	Bit 7 = RESERVED = 0b Bits 6:5 = Multi-SST Operation = 00b (Not supported) Bit 4 = RESERVED = 0b

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			Bits 3:0 = Number of Pixels in Horizontal Pixel Count Overlapping an Adjacent Panel Segment = 0000b
92h	0x84	**Checksum Section**	

The Sink represented by this example structure interprets pixel data as being in the sRGB color space and transfer function when receiving a DisplayPort VSC packet indicting sRGB.

The Sink represented by this example structure interprets pixel data as being in the native color space and transfer function indicated in the Display Parameters Data Block when receiving a DisplayPort VSC packet indicting Custom Color RGB, or when receiving a DisplayPort MSA packet indicating RGB Unspecified.

Background Information

The example structure declares native primaries and native EOTF in the Display Parameters Data Block. However, it is not clear what signaling a DisplayPort Source device should use to access this color space and transfer function. This is clarified by adding the VESA Organization Vendor-specific Data Block to indicate that MSA packet MISC signaling of "RGB Unspecified" is interpreted by the Sink as being in the native color space and EOTF. Additional guidance is included regarding use of the newer DisplayPort VSC packet signaling to specify sRGB, and Custom Color RGB (native primaries and EOTF).

- End of Document -

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VESA STANDARDS CHANGE REQUEST FORM

To be Filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	Color Depth Mode Support and YCbCr 420 clarification
AFFECTED DOCUMENT:	DisplayID ver 2.0
REVISION CATEGORY:	Refer to VP235H Appendix A; will be subject to Task Group review
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Syed Athar Hussain, AMD

SCR REVISION HISTORY	
(DATE)	**(CHANGE)**
04/09/2019	Draft 1
04/22/2019	Draft 2 update based on the task group feedback

(add more rows as needed)

To be Filled in by VESA Office:

VESA SCR NUMBER:	(To be assigned by VESA office)
SCR ENTRY DATE:	04/09/2019

To be Filled in by Task Group or VESA Office

SCR ADOPTED, REJECTED, or otherwise DISPOSITIONED for other action	SCR is (adopted) or (rejected) or (Dispositioned for other action) If rejected, explain reason for acceptation or rejection If dispositioned, explain action or plan for action (such as including in future draft specification revision, or re-visiting at future date, or other)
DATE SCR ADOPTED or REJECTED or DISPOSITIONED	07/05/2019 - Adopted

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Summary of the Proposed Change(s)

The SCR clarifies mode support rules for various encoding formats with and without certain CTA blocks that are used to limit the mode support to specific VIC, for example YCbCr420 capability map data block and YCbCr420 video data block.

IPR (Intellectual Property Rights) declaration, if any

None

Benefits as a Result of the Changes

Clarification on mode support behavior.

Assessment of the Impact

Provide details.

Analysis of the Device Hardware Implication

No hardware impact is expected.

Analysis of the Device Software Implications

SW and FW may need to be updated to apply the clarification in this document.

Analysis of the Compliance Test & Interop Implications

Expected impact on compliance specially for case of YCbCr 420 mode support behavior with DisplayID.

New Referenced Documents Resulting from Change

DisplayID Ver 2.0.

Attachments

None

Proposed Document Change(s) or Addition(s)

Background Information

4.5.1 Supported Interface Color Depth-related Fields

The four Supported Interface Color Depth-related fields – one byte each for RGB, YCbCr 4:4:4, YCbCr 4:2:2, and YCbCr 4:2:0 pixel encodings, as listed in Table 4-24 – define the supported color depths for corresponding ~~that particular~~ pixel encoding.

Definition of "Supported" and "No support indicated" is as follows:

• **Supported** – Color depth is supported for all supported ~~DisplayID exposed~~ timings. Supported timing includes all Display-ID exposed timings (that is timing exposed using DisplayID timing types and CTA VICs) unless a CTA block is exposed that restrict the support to a limited set of timing for particular pixel encoding, for example, YCbCr420 capability map data block and/or YCbCr420 video data block. If those CTA blocks are exposed, then the support for that pixel encoding would be limited to the timings exposed in the restricted set only.

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•**No support indicated** – Sink device has not explicitly reported a color-depth capability in the Display Interface Features data block; thus, the Source device shall rely on other DisplayID data blocks, –or– an interface-specific rule, to determine color-depth support. Examples of other data blocks that may report color-depth capability include ~~the CTA YCbCr 4:2:0,~~ HDMI vendor-specific, and HDMI Forum Vendor-specific data blocks.

Table 4-24: Supported Interface Color Depth-related Field Formats

Offset	Bit #	Description/Format
03h	**Supported Interface Color Depth for RGB Encoding**	
	0	**6 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	1	**8 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	2	**10 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	3	**12 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	4	**14 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	5	**16 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	7:6	**RESERVED** Cleared to all 0s.

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Table 4-24: Supported Interface Color Depth-related Field Formats (Continued)

Offset	Bit #	Description/Format
04h	**Supported Interface Color Depth for YCbCr 4:4:4 Encoding**	
	0	**6 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	1	**8 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	2	**10 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	3	**12 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	4	**14 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	5	**16 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	7:6	**RESERVED** Cleared to all 0s.

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Table 4-24: Supported Interface Color Depth-related Field Formats (Continued)

Offset	Bit #	Description/Format
05h	**Supported Interface Color Depth for YCbCr 4:2:2 Encoding**	
	0	**8 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	1	**10 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	2	**12 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	3	**14 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	4	**16 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	7:5	**RESERVED** Cleared to all 0s.
06h	**Supported Interface Color Depth for YCbCr 4:2:0 Encoding**	
	0	**8 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	1	**10 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	2	**12 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	3	**14 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	4	**16 bits per Primary Color** 0 = No support indicated. 1 = Supported.
	7:5	**RESERVED** Cleared to all 0s.

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4.5.2 Minimum Pixel Rate at Which YCbCr420 Encoding Is Supported

The Minimum Pixel Rate at Which YCbCr 4:2:0 Encoding Is Supported field indicates the minimum pixel rate at which the Sink device shall support YCbCr 4:2:0 encoding, as indicated in the Supported Interface Color Depth for YCbCr 4:2:0 Encoding field (offset 06h). The Source device shall use YCbCr 4:2:0 encoding only for supported ~~DisplayID-exposed~~ modes that meet or exceed the listed pixel rate. A value of 00h in this field shall indicate that the Sink device can support YCbCr 4:2:0 encoding at all supported ~~DisplayID-exposed~~ modes. Note that if, YCbCr420 capability map data block and YCbCr420 video data block is used to limit the YCbCr 420 timing support then the field shall be set to 00h and support for 420 pixel encoding is limited to the timings exposed in the restricted set exposed in the CTA data block. For further details of supported mode rules please refer to section 4.5.1.

Table 4-25 defines the byte format.

Table 4-25: Minimum Pixel Rate at Which YCbCr 4:2:0 Encoding Is Supported Field Format

Offset	Bit #	Description
07h	7:0	**Minimum Pixel Rate at Which YCbCr 4:2:0 Encoding Is Supported** YCbCr 4:2:0 Minimum Pixel Rate Requirement = 74.25MP/s × field value 00h = Sink device can support YCbCr 4:2:0 encoding at all DisplayID-exposed modes.

- End of Document -

VESA STANDARDS CHANGE REQUEST FORM

To be Filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	DSC pass-through timing support
AFFECTED DOCUMENT:	DisplayID ver 2.0
REVISION CATEGORY:	Refer to VP235H Appendix A; will be subject to Task Group review
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Syed Athar Hussain, AMD

SCR REVISION HISTORY	
(DATE)	**(CHANGE)**
07/16/2019	Initial Draft, rev1
07/30/2019	Rev2, Incorporated task group feedback on explicit tie in between passthrough timing and the VESA VSDB fields.

(add more rows as needed)

To be Filled in by VESA Office:

VESA SCR NUMBER:	(To be assigned by VESA office)
SCR ENTRY DATE:	07/16/2019

To be Filled in by Task Group or VESA Office

SCR ADOPTED, REJECTED, or otherwise DISPOSITIONED for other action	SCR is (adopted) or (rejected) or (Dispositioned for other action) If rejected, explain reason for acceptation or rejection If dispositioned, explain action or plan for action (such as including in future draft specification revision, or re-visiting at future date, or other)
DATE SCR ADOPTED or REJECTED or DISPOSITIONED	10/04/2019 - Adopted

Summary of the Proposed Change(s)

The SCR extends exiting data blocks to support DSC pass-through for some exposed timings. A new version of the Type VII timing data block with a flag to indicate pass-through timing support is added along with new bytes in VESA vendor specific data block to expose the DSC integer and fractional target bit depth.

IPR (Intellectual Property Rights) declaration, if any

None

Benefits as a Result of the Changes

DSC pass-through timing support.

Assessment of the Impact

SW/FW update on source and sinks side.

Analysis of the Device Hardware Implication

No hardware impact is expected.

Analysis of the Device Software Implications

SW and FW may need to be updated to support the new field described in the document.

Analysis of the Compliance Test & Interop Implications

Currently no compliance exits, if required a new DSC pass-through compliance support can be verified with this framework.

New Referenced Documents Resulting from Change

DisplayID Ver 2.0.

Attachments

None

Proposed Document Change(s) or Addition(s)

Background Information

4.3.1 Type VII Timing – Detailed Timing Data Block

Table 4-17: Type VII Timing – Detailed Timing Data Block Format

Offset	Bit #	Description/Format/Priority	Tag
00h	7:0	**Type VII Timing – Detailed Timing Data Block** Value is 22h.	✔
01h	**Block Revision and Other Data**		
	2:0	**Block Revision** Revision values range from 0 through 7. 000b = Revision 0 (default). **001b = Revision 1 (redefined (bit 3 of byte 1) as a flag indicating DSC pass-through timing support for all listed timing in the block)**	
	3	**For Block Revision 0** Bit 3 shall be 0. **For Block Revision 1** Value 0 = Same as revision 0 (standard timing support declaration ~~only~~ as original specification) Value 1 = Indicates listed timing descriptors in the data block are supported with DSC pass-through with DSC RGB encoding and specific target DSC bits per pixel only. ~~Pass-through bpp applies as declared in VESA organization vendor-specific data block.~~ For DisplayPort, VESA organization vendor-specific data block shall exists with relevant pass-through integer and fractional DSC bits per pixel declaration if this bit is set; otherwise source shall disregard the timings. ~~DP Tx and Rx shall follow DisplayPort specification rules pertaining to other DSC capability support.~~	
	7:4	**RESERVED** Cleared to all 0s.	
02h	7:0	**Number of Payload Bytes in Block** Number of bytes within the data block range from 20 through 240 ($1 \leq N \leq 12$). Values are based on $N \times 20$ – 14h, 28h, 3Ch, and so forth, through F0h. All other values are RESERVED.	
03h – 16h	159:0	**First Type VII Detailed Timing Descriptor** 20-byte descriptor. Priority 1.	
17h – 2Ah	159:0	**Second Type VII Detailed Timing Descriptor** 20-byte descriptor, if present. Priority 2.	
…		…	

B VESA Organization Vendor-specific Data Block

Table B-1: VESA Organization Vendor-specific Data Block (Continued)

Offset	Bit #	Description/Format	Tag
07h	5:0	**Pass through timing's integer target DSC bits per pixel** Integer portion of target DSC bits per pixel for pass-through timing descriptors declared in Type VII timing data block with revision 1 and higher and byte 1h and bit 3 set to value 1 of the timing data block. DP Tx and Rx shall follow DisplayPort specification rules pertaining to other DSC capability support. Valid value is 0 and 8 through 16 integer target DSC bits per pixel, all other values are reserved and shall not be used.	
	7:6	**RESERVED** Shall be set to 0.	
08h	3:0	**Pass through timing's fractional target DSC bits per pixel** Fractional portion of target DSC bits per pixel for pass-through timing descriptors declared in Type VII timing data block with revision 1 and higher and byte 1h and bit 3 set to value 1 of the timing data block. DP Tx and Rx shall follow DisplayPort specification rules pertaining to other DSC capability support. Valid value range is 0 through 15 corresponding to fraction DSC bit per pixel of (1/16) * value. Note value of 0 indicates no fractional DSC bits per pixel exists.	
	7:4	**RESERVED** Shall be set to 0.	

- End of Document -

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VESA STANDARDS CHANGE REQUEST FORM

To be Filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	Maximum Vertical Refresh Rate field extension of Dynamic Video Timing Range Limit Data Block
AFFECTED DOCUMENT:	DisplayID ver 2.0
REVISION CATEGORY:	Refer to VP235H Appendix A; will be subject to Task Group review
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Syed Athar Hussain, AMD

SCR REVISION HISTORY	
(DATE)	**(CHANGE)**
12/17/2019	Initial Draft, rev1

(add more rows as needed)

To be Filled in by VESA Office:

VESA SCR NUMBER:	(To be assigned by VESA office)
SCR ENTRY DATE:	12/17/2019

To be Filled in by Task Group or VESA Office

SCR ADOPTED, REJECTED, or otherwise DISPOSITIONED for other action	SCR is (adopted) or (rejected) or (Dispositioned for other action) If rejected, explain reason for acceptation or rejection If dispositioned, explain action or plan for action (such as including in future draft specification revision, or re-visiting at future date, or other)
DATE SCR ADOPTED or REJECTED or DISPOSITIONED	03/06/2020 Adopted

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Summary of the Proposed Change(s)

The SCR extends the Maximum Vertical Refresh Rate field of the Dynamic Video Timing Range Limit Data Block to support higher the 255Hz refresh rate. The SCR also clarifies some of the range values of other fields to explicitly call out the min and max possible values.

IPR (Intellectual Property Rights) declaration, if any

None

Benefits as a Result of the Changes

With this change Adaptive-Sync displays with 255Hz and more can enumerate their capability in a standard method.

Assessment of the Impact

SW/FW update on source and sinks side.

Analysis of the Device Hardware Implication

No hardware impact is expected if higher refresh rate timings are supported. If not supported, then source shall not expose the higher refresh rate timing and revert to a lower timing mode.

Analysis of the Device Software Implications

SW and FW may need to be updated to support the new field associated with new data block revision described in the document.

Analysis of the Compliance Test & Interop Implications

Adaptive-Sync related compliance test would need to be updated for source and sink testing of beyond 255Hz.

New Referenced Documents Resulting from Change

DisplayID Ver 2.0 + SCR.

Attachments

None

Proposed Document Change(s) or Addition(s)

Background Information

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Dynamic Video Timing Range Limits Data Block

The Dynamic Video Timing Range Limits data block shall be used to convey the supported timing range of vertical frequencies. Any Sink device that exposes this data block shall indicate support for a seamless, continuous vertical frequency device that can support the seamless timing change for any timing that lies within the specified minimum and maximum pixel rates. Dynamic Video Timing Range Limits is an **optional** data block in Base sections.

Table 4-22 lists the size and order of each field, with all addresses relative to the beginning of the data block.

Table 4-22: Dynamic Video Timing Range Limits Data Block Fields

Offset	Bit #	Description/Format	Flag	Tag
00h	7:0	**Dynamic Video Timing Range Limits Data Block** Value is 25h.		✓
01h	**Block Revision and Other Data**			
	2:0	**Block Revision** Revision ranges from 0 through 7. ~~000b = Revision 0 (default).~~ 001b = Revision 1		
	7:3	**RESERVED** Cleared to all 0s.		
02h	7:0	**Number of Payload Bytes in Block** 09h = Data block is composed of nine payload bytes. All other values are RESERVED.		
05h– 03h	23:0	**Minimum Pixel Clock (in kHz)** ~~Required.~~ The field is required to be populated and specifies the minimum pixel rate above which dynamic video timing changes shall be supported, as per the reported scheme. Range shall be defined as 0.001 through 16,777.216MP/s, where value of 000000h corresponds to 0.001 MP/s and value of FFFFFFh corresponds to 16,777.216MP/s. ~~Value ranges from 000000h through FFFFFFh.~~		

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08h – 06h	23:0	**Maximum Pixel Clock (in kHz)** ~~Required.~~ The field is required to be populated and specifies the maximum pixel rate below which dynamic video timing changes shall be supported, as per the reported scheme. Range shall be defined as 0.001 through 16,777.216MP/s, where value of 000000h corresponds to 0.001 MP/s and value of FFFFFFh corresponds to 16,777.216MP/s. ~~Value ranges from 000000h through FFFFFFh.~~		
09h	7:0	**Minimum Vertical Refresh Rate** ~~Required.~~ The field is required to be populated and specifies the display device's minimum dynamic vertical frequency. Vertical frequency shall be defined as follows: *(pixel_clock/horizontal_total) / vertical_total* Range shall be defined as 0 through 255Hz, where value of 00h corresponds to 0Hz and value of FF corresponds to 255Hz. ~~Value ranges from 00h through FFh.~~		

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Table 4-22: Dynamic Video Timing Range Limits Data Block Fields (Continued)

Offset	Bit #	Description/Format	Flag	Tag
0Ah	7:0	**Maximum Vertical Refresh Rate (8 least significant bits)** ~~Required.~~ The field is required to be populated and specifies the display device's maximum dynamic vertical frequency. In version 0 of the data block, the "**Maximum Vertical Refresh Rate"** field was 8 bits wide and the range is defined as 0 through 255Hz, where value of 00h corresponds to 0Hz and value of FF corresponds to 255Hz. In version 1 of the data block, the "**Maximum Vertical Refresh Rate"** field is 10 bits wide and the range is defined as 0 through 1023Hz, where value of 00h corresponds to 0Hz and value of 3FF corresponds to 1023Hz. The 8 least significant bits are defined in offset 0Ah and remaining 2 bits are defined in offset 0Bh bit 0 and 1.		
0Bh	**Dynamic Video Timing Range Related and Support Flags**			
	1:0	**Maximum Vertical Refresh Rate most significant bits 9 and 10** In version 0 of the data block the field is set to zero. In version 1 of the data block the bit 0 correspond to the bit 9 and bit 1 correspond to the bit 10 of the "Maximum Vertical Refresh Rate" to all the range to be extended from a max 255 Hz to 1023 Hz.		
	6:2	**RESERVED** Cleared to all 0s.		
	7	**Seamless Dynamic Video Timing Support** 0 = Seamless Dynamic Video Timing change shall not be supported with a fixed horizontal pixel rate and dynamic vertical blanking. 1= Seamless Dynamic Video Timing change shall be supported with a fixed horizontal pixel rate and dynamic vertical blanking.	✓	

- End of Document -

VESA STANDARDS CHANGE REQUEST FORM

To be Filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	Updated type VII timing data block to add future extensibility and accommodate CTA request to explicitly indicate YCC 420 support in each timing descriptor.
AFFECTED DOCUMENT:	DisplayID ver ~~2.1~~ 2.0
REVISION CATEGORY:	Refer to VP235H Appendix A; will be subject to Task Group review
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Syed Athar Hussain, AMD

SCR REVISION HISTORY	
(DATE)	**(CHANGE)**
9/16/2020	Initial Draft, rev1
11/11/2020	Draft rev 2; Address all GMR comments
11/20/2020	Merged changes with previous Type VII SCR

(add more rows as needed)

To be Filled in by VESA Office:

VESA SCR NUMBER:	(To be assigned by VESA office)
SCR ENTRY DATE:	09/16/20

To be Filled in by Task Group or VESA Office

SCR ADOPTED, REJECTED, or otherwise DISPOSITIONED for other action	SCR is (adopted) or (rejected) or (Dispositioned for other action) If rejected, explain reason for acceptation or rejection If dispositioned, explain action or plan for action (such as including in future draft specification revision, or re-visiting at future date, or other)
DATE SCR ADOPTED or REJECTED or DISPOSITIONED	ADOPTED 11/30/2020

Summary of the Proposed Change(s)

A new version of type VII timing data block is defined to add future extensibility and accommodate CTA request to explicitly indicate YCC 420 support in each timing descriptor.

IPR (Intellectual Property Rights) declaration, if any

None

Benefits as a Result of the Changes

With the proposed updates, following benefit can be realized:

1. Future descriptor can be defined with more than 20 Byte per descriptor without creating a new timing type.
2. Added Explicit YCC 420 per timing descriptor as per CTA request.

Assessment of the Impact

SW/FW update on source and sinks side.

Analysis of the Device Hardware Implication

No hardware impact is expected.

Analysis of the Device Software Implications

SW and FW may need to be updated to support the new data block revision described in the document. Also implementation shall accommodate future extensibility.

Analysis of the Compliance Test & Interop Implications

DisplayID compliance would need be updated to ensure proper source and sink support of the new capability.

New Referenced Documents Resulting from Change

DisplayID ver ~~2.1~~ 2.0

Attachments

None

Proposed Document Change(s) or Addition(s)

Background Information

4.3.1 Type VII Timing – Detailed Timing Data Block

The Type VII Detailed Timing data block shall be composed of single or multiple 20-byte Detailed Timing descriptors. Type VII is a **mandatory** data block for product primary use case as a display and **optional** otherwise.

The *DisplayID v2.0* definition of the Type VII Detailed Timing data block is similar to the *DisplayID v1.3* Type I Detailed Timing data block except for Byte 0, 1, 2 of the descriptor fields, which carry the pixel clock information. With Type VII, higher-precision pixel clock support can be exposed. The field's range allows pixel clock definition from 0.001 to 16,777.216MP/s.

Table 4-17 and Table 4-18 list the data block's structure and descriptor formats, respectively.

Table 4-17: Type VII Timing – Detailed Timing Data Block Format

Offset	Bit #	Description/Format/Priority
00h	7:0	**Type VII Timing – Detailed Timing Data Block** Value is 22h.
01h	**Block Revision and Other Data**	
	2:0	**Block Revision** Revision values range from 0 through 7. 000b = Revision 0 ~~(default).~~ 001b = Revision 1 (redefined bit 3 of byte 1 as a flag indicating DSC pass-through timing support for all listed timings in the block) 010b = Revision 2 (recommended for new designs). All other values are RESERVED.
	3	**RESERVED** **For Block Revision 0** Bit 3 shall be 0. **For Block Revision 1 and higher** Value 0 = Same as revision 0 (standard timing support declaration as original specification) Value 1 = Indicates listed timing descriptors in the data block are supported with DSC pass-through with DSC RGB encoding and specific target DSC bits per pixel only. For DisplayPort, VESA organization vendor-specific data block shall exist with relevant pass-through integer and fractional DSC bits per pixel declaration if this bit is set; otherwise source shall disregard the timings. Usage of the field with CTA-861: When the Type VII Timing Data Block is packed in a CTA Data Block, bit 3 shall be set to 0.
	6:4	**Number of Payload Bytes (M) in a Timing Descriptor** Where M (Bytes) = 20 (Initial descriptor size) + value of the field 000b = 20+0 bytes per descriptor (Defined as part of Type VII, revision 1) All other values are RESERVED.
	7	**RESERVED** Cleared to all 0s.

02h	7:0	**Number of Payload Bytes in Block** ~~Number of bytes within the block range from 20 through 240 ($1 \leq N \leq 12$).~~ ~~Values are based on $N \times 20$ – 14h, 28h, 3Ch, and so forth, through F0h.~~ Number of payload bytes within the block is based on # of descriptor (N) x Size of each descriptor (M) bytes. Note: value of M is based on "Number of Payload Bytes (M) in a Timing Descriptor" field. Source implementation shall accommodate varying field for future extensibility. All other values are RESERVED. Usage with CTA-861: When the Type VII Timing Data Block is packed in a CTA Data Block, which separately indicates the length of the CTA Data Block payload, the "Number of Payload Bytes in Block" field shall not be present and the size of the Type VII Timing Data Block is reduced by 1 byte.
03h – 03h+M-1	(M*8-1):0	**First Type VII Detailed Timing Descriptor** M-byte descriptor. Priority 1.
03h+M – 03h+2M-1	(M*8-1):0	**Second Type VII Detailed Timing Descriptor** M-byte descriptor, if present. Priority 2.
…		…
03h+(N-1)*M – 03h+(N*M)-1	(M*8-1):0	**Nth Type VII Detailed Timing Descriptor** M-byte descriptor, if present. Priority N.

Table 4-18: Type VII Detailed Timing Descriptor Format

Byte #	Bit #	Description/Format
2:0	**Pixel Clock ÷ 1000** Range is defined as 0.001 through 16,777.216MP/s.	
0	7:0	**Pixel Clock ÷ 1000 Low Bits 7:0** Values range from 00h through FFh.
1	15:8	**Pixel Clock ÷ 1000 Middle Bits 15:8** Values range from 00h through FFh.
2	23:16	**Pixel Clock ÷ 1000 High Bits 23:16** Values range from 00h through FFh.

Table 4-18: Type VII Detailed Timing Descriptor Format (Continued)

Byte #	Bit #	Description/Format
3	**Timing Options**	
	3:0	**Aspect Ratio** 0h = 1:1 4h = 16:9 1h = 5:4 5h = 16:10 2h = 4:3 6h = 64:27 3h = 15:9 7h = 256:135 8h = Aspect ratio shall be calculated using the Horizontal Active Image Pixels and Vertical Active Image Lines fields (Bytes 5:4 and 13:12, respectively). 9h through Fh = RESERVED.
	4	**Interface Frame Scanning Type** 0 = Progressive scan frame. 1 = Interlaced scan frame.
	6:5	**3D Stereo Support** 00b = This timing shall always be displayed in mono (no stereo). 01b = This timing shall always be displayed in stereo. 10b = This timing shall be displayed in mono or stereo, depending on a user action (such as wearing stereo glasses). 11b = RESERVED
	7	**Block Revision 0 and 1:** **Preferred Detailed Timing** 1 = Preferred Detailed timing. **Block Revision 2 and later:** **YCC 420 Support** 0 = No explicit YCC 420 support indicated for current mode (other blocks may be utilized to indicate support) 1 = Explicit YCC 420 support indicated for current timing
5:4	**Horizontal Active Image Pixels** Pixel quantities range from 1 through 65,536.	
4	7:0	**Horizontal Active Image Pixels Low Bits 7:0** Values range from 00h through FFh.
5	15:8	**Horizontal Active Image Pixels High Bits 15:8** Values range from 00h through FFh.
7:6	**Horizontal Blank Pixels** Pixel quantities range from 1 through 65,536.	
6	7:0	**Horizontal Blank Pixels Low Bits 7:0** Values range from 00h through FFh.
7	15:8	**Horizontal Blank Pixels High Bits 15:8** Values range from 00h through FFh.

Table 4-18: Type VII Detailed Timing Descriptor Format (Continued)

Byte #	Bit #	Description/Format
9:8	**Horizontal Offset (Front Porch)** Pixel quantities range from 1 through 32,768.	
8	7:0	**Horizontal Offset (Front Porch) Low Bits 7:0** Values range from 00h through FFh.
9	14:8	**Horizontal Offset (Front Porch) High Bits 14:8** Values range from 00h through 7Fh.
	15	**Horizontal Sync Polarity** 0 = Negative. 1 = Positive.
11:10	**Horizontal Sync Width** Pixel quantities range from 1 through 65,536.	
6	7:0	**Horizontal Sync Width Low Bits 7:0** Values range from 00h through FFh.
7	15:8	**Horizontal Sync Width High Bits 15:8** Values range from 00h through FFh.
13:12	**Vertical Active Image Lines** Line quantities range from 1 through 65,536.	
4	7:0	**Vertical Active Image Lines Low Bits 7:0** Values range from 00h through FFh.
5	15:8	**Vertical Active Image Lines High Bits 15:8** Values range from 00h through FFh.
15:14	**Vertical Blank Lines** Line quantities range from 1 through 65,536.	
14	7:0	**Vertical Blank Lines Low Bits 7:0** Values range from 00h through FFh.
15	15:8	**Vertical Blank Lines High Bits 15:8** Values range from 00h through FFh.
17:16	**Vertical Sync Offset (Front Porch)** Line quantities range from 1 through 32,768.	
16	7:0	**Vertical Sync Offset (Front Porch) Low Bits 7:0** Values range from 00h through FFh.
17	14:8	**Vertical Sync Offset (Front Porch) High Bits 14:8** Values range from 00h through 7Fh.
	15	**Vertical Sync Polarity** 0 = Negative. 1 = Positive.

Table 4-18: Type VII Detailed Timing Descriptor Format (Continued)

Byte #	Bit #	Description/Format
19:18	**Vertical Sync Width** Line quantities range from 1 through 65,536.	
18	7:0	**Vertical Sync Width Low Bits 7:0** Values range from 00h through FFh.
19	15:8	**Vertical Sync Width High Bits 15:8** Values range from 00h through FFh.

- End of Document -

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VESA STANDARDS CHANGE REQUEST FORM

To be Filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	Created a new type X timing data block to support CVT1.3 and support refresh rate field up to 1024 Hz.
AFFECTED DOCUMENT:	DisplayID ver ~~2.1~~ 2.0
REVISION CATEGORY:	Refer to VP235H Appendix A; will be subject to Task Group review
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Syed Athar Hussain, AMD

SCR REVISION HISTORY	
(DATE)	**(CHANGE)**
8/19/20	Initial Draft, rev1
8/24/20	Rev2, incorporate feedback from task group to allow 6-byte descriptor along with 7-byte descriptor.
9/8/20	Rev3, added provision to expose YCC420 support for each timing descriptor to accommodate CTA request.
11/06/20	Rev4, added provision to not include "Number of Payload Bytes in Block" byte when the block is packed in CTA block

(add more rows as needed)

To be Filled in by VESA Office:

VESA SCR NUMBER:	(To be assigned by VESA office)
SCR ENTRY DATE:	08/24/20

To be Filled in by Task Group or VESA Office

SCR ADOPTED, REJECTED, or otherwise DISPOSITIONED for other action	SCR is (adopted) or (rejected) or (Dispositioned for other action) If rejected, explain reason for acceptation or rejection If dispositioned, explain action or plan for action (such as including in future draft specification revision, or re-visiting at future date, or other)
DATE SCR ADOPTED or REJECTED or DISPOSITIONED	ADOPTED 11/30/2020

Summary of the Proposed Change(s)

The SCR creates the type X data block to add CVT 1.3 support and extend the Refresh Rate field to support up to 1024 Hz.

IPR (Intellectual Property Rights) declaration, if any

None

Benefits as a Result of the Changes

With the proposed updates, following benefit can be realized:

1. CVT 1.3 reduced blanking version 3 timing can be exposed using type X Timing data block revision 0. Earlier type IX was limited to CVT 1.2 specification.
2. Type X data block version 0 will support up to 1024 Hz timing modes. Earlier type IX data block was limited to 256Hz timing.
3. Added Explicit YCC 420 per timing descriptor as per CTA request.
4. Added provision to not include the "Number of Payload Bytes in Block" byte when the Type X data block is packed in CTA block.

Assessment of the Impact

SW/FW update on source and sinks side.

Analysis of the Device Hardware Implication

No hardware impact is expected if CVT 1.3 timing rules are supported.

No hardware impact if higher refresh rate timings are supported by sink and source. If not supported, then sink shall not expose the higher refresh rate timing.

Analysis of the Device Software Implications

SW and FW may need to be updated to support the new field associated with new data block revision described in the document.

Analysis of the Compliance Test & Interop Implications

Since CVT 1.3 is being mandated for Adaptive-Sync. Adaptive-Sync related compliance test would need to be updated for source and sink testing.

New Referenced Documents Resulting from Change

DisplayID ver ~~2.1~~2.0

Attachments

None

Proposed Document Change(s) or Addition(s)

Background Information

4.3.3 Type X Timing – Formula-based Data Block

The Type X Formula-based data block provides a means for a Sink device to expose timing support that is derived from an industry-standard formula. The revision 0 block supports *CVT v1.3* timing with standard blanking, and reduced blanking version 1, version 2 and version 3. The 7-byte descriptor variant of the block also supports refresh rate field that extends to 1024 Hz. Note, the type X data block is a superset of old type IX data block and can accommodate all of the type IX capability exposure using a 6-byte descriptor variant, it is strongly recommended that new sink move to utilizing the Type X data block.

Sink may list its timing priority in the order of timing exposure as a hint, however, note source may not utilize sink exposed priority and instead implement a source implementation specific policy based on native resolution of the display and link capability.

Type X Timing is an **optional** data block.

Table 4-XX and Table 4-XY list the data block's structure and descriptor formats, respectively.

Table 4-XX: Type X Timing – Formula-based Data Block Format

Offset	Bit #	Description/Format/Priority
00h	7:0	**Type X Timing – Formula-based Timing Data Block** Value is 2Ah.
01h	**Block Revision and Other Data**	
	2:0	**Block Revision** Revision values range from 0 through 7. 000b = Revision 0. All other values are RESERVED.
	3	Cleared to 0.
	6:4	**Number of Payload Bytes (M) in a Timing Descriptor** Where M (Bytes) = 6 (Initial descriptor size) + value of the field 000b = 6+0 bytes per descriptor (Defined as part of Type X, revision 0) 001b = 6+1 bytes per descriptor (Defined as part of Type X, revision 0) All other values are RESERVED.
	7	**RESERVED** Cleared to 0.

02h	7:0	**Number of Payload Bytes in Block** Number of payload bytes within the block is based on # of descriptor (N) x Size of each descriptor (M) bytes. Note: value of M is based on "**Number of Payload Bytes (M) in a Timing Descriptor".** Source implementation shall accommodate varying field for future extensibility. All other values are RESERVED. Usage with CTA-861: When the Type X Timing Data Block is packed in a CTA Data Block, which separately indicates the length of the CTA Data Block payload, the "Number of Payload Bytes in Block" field shall not be present and the size of the Type X Timing Data Block is reduced by 1 byte.
03h – 03h+M-1	(M*8-1):0	**First Type X Formula Timing Descriptor** M-byte descriptor. Priority 1.
03h+M – 03h+2M-1	(M*8-1):0	**Second Type X Formula Timing Descriptor** M-byte descriptor if present. Priority 2.
…		…
03h+(N-1)*M – 03h+(N*M)-1	(M*8-1):0	**Nth Type X Formula Timing Descriptor** M-byte descriptor. Priority N.

Table 4-XY: Type X Formula-based Timing Descriptor Format

Byte #	Bit #	Description/Format	6-Byte	7-Byte
0	**Timing Options**		✔	✔
	2:0	**Timing Formula/Algorithm** 000b = *CVT* standard blanking. 001b = *CVT* reduced blanking version 1. 010b = *CVT* reduced blanking version 2. 011b = *CVT* reduced blanking version 3. All other values are RESERVED.		
	3	**RESERVED** Cleared to 0.	✔	✔
	4	The definition of bit 4 meaning is dependent on Timing Formula/Algorithm field. If Timing Formula/Algorithm field is set to 010b (i.e. reduced blanking version 2) then the bit has following meaning: **CVT version 2: Video Refresh Rate Option** 0 = Refresh rate × (1000 / 1001) variant of timing is not supported. 1 = Refresh rate × (1000 / 1001) variant of timing is also supported, along with base Refresh rate. If Timing Formula/Algorithm field is set to 011b (i.e. reduced blanking version 3) then the bit has following meaning: **CVT version 3: Horizontal Blank Option** 0 = Horizontal Blank of 80 Pixels. 1 = Horizontal Blank of 160 Pixels. For all other values of Timing Formula/Algorithm field, the bit shall be cleared to 0b.	✔	✔
	6:5	**3D Stereo Support** 00b = This timing shall always be displayed in mono (no stereo). 01b = This timing shall always be displayed in stereo. 10b = This timing shall be displayed in mono or stereo, depending on a user action (such as wearing stereo glasses). 11b = RESERVED.	✔	✔
	7	**YCC 420 Support** 0 = No explicit YCC 420 support for current mode (other blocks may be utilized to indicate support) 1 = YCC 420 supported for current timing	✔	✔
2:1	**Horizontal Active Image Pixels** Pixel quantities range from 1 through 65,536.		✔	✔
1	7:0	**Horizontal Active Image Pixels Low Bits 7:0** Values range from 00h through FFh.	✔	✔
2	15:8	**Horizontal Active Image Pixels High Bits 15:8** Values range from 00h through FFh.	✔	✔

4:3	**Vertical Active Image Lines** Line quantities range from 1 through 65,536.		✔	✔
3	7:0	**Vertical Active Image Lines Low Bits 7:0** Values range from 00h through FFh.	✔	✔
4	15:8	**Vertical Active Image Lines High Bits 15:8** Values range from 00h through FFh.	✔	✔
5	7:0	**Refresh Rate Low Bits 7:0** Least significant 8 bits of Refresh Rate field [bit0 …bit7]. Note: 6 Byte Type X Formula-based Timing Descriptor end at this offset and this limits the Refresh Rate field to 256Hz. As value of 0 correspond to 1Hz.	✔	✔
6	1:0	**Refresh Rate High Bits 9:8** Note: Defined only for 7 Byte Type X Formula-based Timing Descriptor. The value of the Refresh Rate field comprises of byte 5 bits [0:7] and byte 6 bits [0:1] to make up the 10 bits Refresh rate field. Refresh Rate has a range of 1 – 1024Hz, with value of 0 corresponds to 1 Hz an 3FFh corresponding to 1024 Hz.		✔
	7:2	**RESERVED** **Cleared to 0.**		✔

- End of Document -

VESA STANDARDS CHANGE REQUEST FORM

To be Filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	Dark Theme preference with exposure of optional Display Device Theme Preference information.
AFFECTED DOCUMENT:	DisplayID ver 2.0
REVISION CATEGORY:	Refer to VP235H Appendix A; will be subject to Task Group review
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Syed Athar Hussain, AMD

SCR REVISION HISTORY	
(DATE)	**(CHANGE)**
7/29/2020	Initial Draft, rev1
11/17/2020	Draft rev2, update based on GMR comment

(add more rows as needed)

To be Filled in by VESA Office:

VESA SCR NUMBER:	(To be assigned by VESA office)
SCR ENTRY DATE:	07/29/2020

To be Filled in by Task Group or VESA Office

SCR ADOPTED, REJECTED, or otherwise DISPOSITIONED for other action	SCR is (adopted) or (rejected) or (Dispositioned for other action) If rejected, explain reason for acceptation or rejection If dispositioned, explain action or plan for action (such as including in future draft specification revision, or re-visiting at future date, or other)
DATE SCR ADOPTED or REJECTED or DISPOSITIONED	ADOPTED 12/14/2020

Summary of the Proposed Change(s)

The SCR adds Display Device Theme Preference ~~Display Device Power Characteristics~~ field in Display Parameter Data block to let sinks optionally expose the theme preference information that source may utilize ~~power characteristics capability~~.

IPR (Intellectual Property Rights) declaration, if any

None

Benefits as a Result of the Changes

The new field may be utilized by source to apply content profile (i.e. dark profile) to optimize for power.

Assessment of the Impact

SW/FW update on source and sinks side.

Analysis of the Device Hardware Implication

No hardware impact is expected.

Analysis of the Device Software Implications

SW and FW may need to be updated to support the new field associated with new data block revision described in the document.

Analysis of the Compliance Test & Interop Implications

Since this is optional no compliance impact is expected

New Referenced Documents Resulting from Change

DisplayID Ver 2.0 + SCR.

Attachments

None

Proposed Document Change(s) or Addition(s)

Background Information

Display Parameters Data Block

The Display Parameters data block is composed of several fields that shall define the monitor's global parameters. No more than one Display Parameters data block shall be exposed within the DisplayID section. Display Parameters is a **mandatory** data block for product primary use case as a display and is **optional** otherwise.

Table 4-7 lists the size and order of each field, with all addresses relative to the beginning of the data block.

Table 4-7: Display Parameters Data Block Fields

Offset	Bit #	Description/Format	Defined in	Tag
00h	7:0	**Display Parameters Data Block** Value is 21h.		✔
01h	**Block Revision and Other Data**			
	2:0	**Block Revision** Revision ranges from 0 through 7. ~~000b = Revision 0 (default).~~ 001b = Revision 1. Added **Display Device Theme Preference** field in byte 1Eh of the data block.		
	6:3	**RESERVED** Cleared to all 0s.		
	7	**Image Size Multiplier** 0 = Horizontal Image Size and Vertical Image Size fields (offsets 03h through 06h) have 0.1-mm precision (default). 1 = Horizontal Image Size and Vertical Image Size fields have 1.0-mm precision.	Section 4.2.1	
02h	7:0	**Number of Payload Bytes in Block** 1Dh = Data block is composed of 29 payload bytes.		
04h– 03h	15:0	**Horizontal Image Size** Descriptor.	Section 4.2.1	
06h– 05h	15:0	**Vertical Image Size** Descriptor.	Section 4.2.1	
08h– 07h	15:0	**Horizontal Pixel Count** Descriptor.	Section 4.2.2	
0Ah– 09h	15:0	**Vertical Pixel Count** Descriptor.	Section 4.2.2	
0Bh	7:0	**Feature Support Flags** Descriptor.	Section 4.2.3	
0Eh – 0Ch	23:0	**Native Color Chromaticity (Primary Color 1 Chromaticity)** Descriptor.	Section 4.2.4	

Table 4-7: Display Parameters Data Block Fields (Continued)

Offset	Bit #	Description/Format	Defined in	Tag
11h – 0Fh	23:0	**Native Color Chromaticity (Primary Color 2 Chromaticity)** Descriptor.	Section 4.2.4	
14h – 12h	23:0	**Native Color Chromaticity (Primary Color 3 Chromaticity)** Descriptor.	Section 4.2.4	
17h – 15h	23:0	**Native Color Chromaticity (White Point Chromaticity)** Descriptor.	Section 4.2.4	
19h – 18h	15:0	**Native Maximum Luminance (Full Coverage)** Descriptor.	Section 4.2.5	
1Bh – 1Ah	15:0	**Native Maximum Luminance (10% Rectangular Coverage)** Descriptor.	Section 4.2.5	
1Dh – 1Ch	15:0	**Native Minimum Luminance** Descriptor.	Section 4.2.5	
1Eh	7:0	**Native Color Depth and Display Device Technology** Descriptor.	Section 4.2.6	
1Fh	7:0	**Native Gamma EOTF** Descriptor.	Section 4.2.7	

Native Color Depth and Display Device Technology Fields

The Native Color Depth and Display Device Technology fields provide information related to the display's native color depth and the technology used by the display device's "glass."

Table 4-14 lists the field formats. In revision 1 of the Display Parameter Data Blocks, Display Device Theme Preference was added to let sinks optionally expose theme preference. For more details on the field, refer to the table below.

Table 4-14: Native Color Depth and Display Device Technology Field Format

Offset	Bit #	Description/Format
1Eh	2:0	**Native Color Depth** Value shall be selected based on the Sink device's ability to use all available bits per component (bpc) for display output rendering. 000b = Not defined. Source device shall apply display interface-specific rules. 001b = 6bpc. 010b = 8bpc. 011b = 10bpc. 100b = 12bpc. 101b = 16bpc. 110b and 111b = RESERVED.

	3	**RESERVED** Cleared to 0.
	6:4	**Display Device Technology** Describes the technology used by the display device's "glass." The Source device may optionally use the information provided by this field as a hint to optimize the Source device's content for the relevant technology. The Sink device shall **not** expect a guaranteed behavior of the Source device based on the information provided in this field. 000b = Technology type is not specified. 001b = Active Matrix LCD technology. 010b = Organic LED technology. All other values are RESERVED.
	7	~~**RESERVED**~~ ~~Cleared to 0.~~ **Display Device Theme Preference** The Source device may optionally use the information provided by this field to optimize source content according to the preference expressed by the Sink device. The Sink device shall not expect any guaranteed behavior of the Source device based on the information provided in this field. 0b = No Preference 1b = Dark Theme Preferred (lower average frame brightness is desired) ~~**Display Device Power Characteristics**~~ ~~The field describes specific power characteristics of the display device. The source device may optionally use the information provided by this field to optimize the source content (i.e. apply dark theme content profile). The Sink device shall not expect a guaranteed behavior of the Source device based on the information provided in this field.~~ ~~0b = Not Defined~~ ~~1b = Lower average frame brightness leads to lower display power consumption~~

- End of Document -

VESA STANDARDS CHANGE REQUEST FORM

To be Filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	Updated Type VIII Timing data block to accommodate CTA request to explicitly indicate YCC 420 support.
AFFECTED DOCUMENT:	DisplayID ver 2.0
REVISION CATEGORY:	Refer to VP235H Appendix A; will be subject to Task Group review
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Bob Ridenour, Apple Inc.

SCR REVISION HISTORY	
(DATE)	**(CHANGE)**
11/9/20	Initial Draft, rev1

(add more rows as needed)

To be Filled in by VESA Office:

VESA SCR NUMBER:	(To be assigned by VESA office)
SCR ENTRY DATE:	11/11/2020

To be Filled in by Task Group or VESA Office

SCR ADOPTED, REJECTED, or otherwise DISPOSITIONED for other action	SCR is (adopted) or (rejected) or (Dispositioned for other action) If rejected, explain reason for acceptation or rejection If dispositioned, explain action or plan for action (such as including in future draft specification revision, or re-visiting at future date, or other)
DATE SCR ADOPTED or REJECTED or DISPOSITIONED	ADOPTED 12/28/2020

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Summary of the Proposed Change(s)

A new version of Type VIII Timing data block is defined to accommodate CTA request to explicitly indicate YCC 420 support and to not include the "Number of Payload Bytes in Block" field when the Type VIII data block is packed in a CTA data block.

IPR (Intellectual Property Rights) declaration, if any

None

Benefits as a Result of the Changes

With the proposed updates, following benefit can be realized:

1. Added Explicit YCC 420 indication as per CTA request.
2. Added provision to not include the "Number of Payload Bytes in Block" field when the Type VIII data block is packed in a CTA data block.

Assessment of the Impact

SW/FW update on source and sinks side.

Analysis of the Device Hardware Implication

No hardware impact is expected.

Analysis of the Device Software Implications

SW and FW may need to be updated to support the new data block revision described in the document.

Analysis of the Compliance Test & Interop Implications

DisplayID compliance would need be updated to ensure proper source and sink support of the new capability.

New Referenced Documents Resulting from Change

DisplayID ver 2.0

Attachments

None

Proposed Document Change(s) or Addition(s)

Background Information

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4.3.2 Type VIII Timing – Enumerated Timing Code Data Block

The Type VIII Enumerated Timing Code data block provides a means for exposing support for timings based on enumerating timing codes within the data block. Timing codes shall be defined by a relevant standards organization such as VESA, the CTA, or HDMI, in DMT Timings (*DMT Standard*), CTA Video Information Code (VIC) Timings, and/or HDMI VIC Timings, respectively. Timing priority shall be in the order listed. A DisplayID section can have multiple Type VIII Timing data blocks. Each Type VIII Timing data block shall include only timings based on a single Timing Code type. Type VIII Enumerated Timing Code is an **optional** data block.

Note: Type VIII Timing data blocks contain CTA VIC timing codes, rather than Short Video Descriptors (SVDs). Native formats shall be inferred from information in the Display Parameters data block, primarily the native format Horizontal Pixel Count and Vertical Pixel Count fields (offsets 08h – 07h and 0Ah – 09h, respectively) of the Display Parameters data block.

Table 4-19 lists the data block's structure.

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Table 4-19: Type VIII Timing – Enumerated Timing Code Data Block Format

Offset	Bit #	Description/Format/Priority
00h	7:0	**Type VIII Timing – Enumerated Timing Code Data Block** Value is 23h.
01h	**Block Revision and Other Data**	
	2:0	**Block Revision** Revision values range from 0 through 7. 000b = Revision 0 ~~(default)~~. 001b = Revision 1 (recommended for new designs).
	3	**Timing Code Size** 0 = 1-byte descriptor timing code 1 = 2-byte descriptor timing code
	4	**RESERVED** Cleared to 0.
	5	**Block Revision 0:** **RESERVED** Cleared to 0. **Block Revision 1 and later:** **YCC 420 Support** 0 = No explicit YCC 420 support indicated for enumerated timing codes (other blocks may be utilized to indicate support) 1 = Explicit YCC 420 support indicated for enumerated timing codes
	7:6	**Timing Code Type** 00b = DMT timing code. 01b = CTA VIC timing code. 10b = HDMI VIC timing code. 11b = RESERVED timing code type

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02h	7:0	**Number of Payload Bytes in Block** Number of payload bytes within the data block ranges from 1 through 248, where a value of 01h corresponds to 1 and a value of F8h corresponds to 248. All other values are RESERVED. Usage with CTA-861: When the Type VIII Timing Data Block is packed in a CTA Data Block, which separately indicates the length of the CTA Data Block payload, the "Number of Payload Bytes in Block" field shall not be present and the size of the Type VIII Timing Data Block is reduced by 1 byte.
03h – 03h+(N-1)	(N*8-1):0	**First N-byte Timing Code** N-byte Descriptor. Priority 1.
03h+N – 03h+2N-1	(N*8-1):0	**Second N-byte Timing Code** N-byte descriptor, if present. Priority 2.
03h+2N – 03h+3N-1	(N*8-1):0	**Third N-byte Timing Code** N-byte descriptor, if present. Priority 3.
…		…

Note: N represents the timing code length that is specified in the Timing Code Size field (i.e., one or two bytes).

- End of Document -

VESA STANDARDS CHANGE REQUEST FORM

To be Filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	Created a new type X timing data block to support CVT ~~1.3~~ RB v3 and support refresh rate field up to 1024 Hz.
AFFECTED DOCUMENT:	DisplayID ver 2.0
REVISION CATEGORY:	Refer to VP235H Appendix A; will be subject to Task Group review
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Syed Athar Hussain, AMD

SCR REVISION HISTORY	
(DATE)	**(CHANGE)**
8/19/20	Initial Draft, rev1
8/24/20	Rev2, incorporate feedback from task group to allow 6-byte descriptor along with 7-byte descriptor.
9/8/20	Rev3, added provision to expose YCC420 support for each timing descriptor to accommodate CTA request.
11/06/20	Rev4, added provision to not include "Number of Payload Bytes in Block" byte when the block is packed in CTA block
2/01/21	Added new flexibility options for additional Horizontal Pixels and Vertical Blank Time to type X
2/10/21	Added tables and examples to clarify the field utilization for added flexibility.
3/31/21	Draft 7. Address GMR comments, changed references from CVT 1.3 to CVT RB v3 and fixed the example in page 4.

(add more rows as needed)

To be Filled in by VESA Office:

VESA SCR NUMBER:	(To be assigned by VESA office)
SCR ENTRY DATE:	03/03/2021

To be Filled in by Task Group or VESA Office

SCR ADOPTED, REJECTED, or otherwise DISPOSITIONED for other action	SCR is (adopted) or (rejected) or (Dispositioned for other action) If rejected, explain reason for acceptation or rejection If dispositioned, explain action or plan for action (such as including in future draft specification revision, or re-visiting at future date, or other)
DATE SCR ADOPTED or REJECTED or DISPOSITIONED	ADOPTED 04/19/2021

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Summary of the Proposed Change(s)

The SCR creates the type X data block to add CVT ~~1.3~~ RB v3 support and extend the Refresh Rate field to support up to 1024 Hz.

IPR (Intellectual Property Rights) declaration, if any

None

Benefits as a Result of the Changes

With the proposed updates, following benefit can be realized:

1. CVT ~~1.3~~ RB v3 reduced blanking version 3 timing can be exposed using type X Timing data block revision 0. Earlier type IX was limited to CVT 1.2 specification.
2. Type X data block version 0 will support up to 1024 Hz timing modes. Earlier type IX data block was limited to 256Hz timing.
3. Added Explicit YCC 420 per timing descriptor as per CTA request.
4. Added provision to not include the "Number of Payload Bytes in Block" byte when the Type X data block is packed in CTA block.

Assessment of the Impact

SW/FW update on source and sinks side.

Analysis of the Device Hardware Implication

No hardware impact is expected if CVT ~~1.3~~ RB v3 timing rules are supported.

No hardware impact if higher refresh rate timings are supported by sink and source. If not supported, then sink shall not expose the higher refresh rate timing.

Analysis of the Device Software Implications

SW and FW may need to be updated to support the new field associated with new data block revision described in the document.

Analysis of the Compliance Test & Interop Implications

Since CVT ~~1.3~~ RB v3 is being mandated for Adaptive-Sync. Adaptive-Sync related compliance test would need to be updated for source and sink testing.

New Referenced Documents Resulting from Change

DisplayID ver ~~2.1~~2.0

Attachments

None

Proposed Document Change(s) or Addition(s)

Background Information

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4.3.3 Type X Timing – Formula-based Data Block

The Type X Formula-based data block provides a means for a Sink device to expose timing support that is derived from an industry-standard formula. The revision 0 block supports *CVT* ~~*v1.3*~~ RB v3 timing with standard blanking, and reduced blanking version 1, version 2 and version 3. The 7-byte descriptor variant of the block also supports refresh rate field that extends to 1024 Hz. Note, the type X data block is a superset of old type IX data block and can accommodate all of the type IX capability exposure using a 6-byte descriptor variant, it is strongly recommended that new sink move to utilizing the Type X data block.

Sink may list its timing priority in the order of timing exposure as a hint, however, note source may not utilize sink exposed priority and instead implement a source implementation specific policy based on native resolution of the display and link capability.

Type X Timing is an **optional** data block.

Table 4-XX and Table 4-XY list the data block's structure and descriptor formats, respectively.

Table 4-XX: Type X Timing – Formula-based Data Block Format

Offset	Bit #	Description/Format/Priority
00h	7:0	**Type X Timing – Formula-based Timing Data Block** Value is 2Ah.
01h	**Block Revision and Other Data**	
	2:0	**Block Revision** Revision values range from 0 through 7. 000b = Revision 0. 001b = Revision 1. All other values are RESERVED.
	3	Cleared to 0.
	6:4	**Number of Payload Bytes (M) in a Timing Descriptor** Where M (Bytes) = 6 (Initial descriptor size) + value of the field 000b = 6+0 bytes per descriptor (Defined as part of Type X, revision 0) 001b = 6+1 bytes per descriptor (Defined as part of Type X, revision 0) All other values are RESERVED.
	7	**RESERVED** Cleared to 0.

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02h	7:0	**Number of Payload Bytes in Block** Number of payload bytes within the block is based on # of descriptor (N) x Size of each descriptor (M) bytes. Note: value of M is based on "**Number of Payload Bytes (M) in a Timing Descriptor".** Source implementation shall accommodate varying field for future extensibility. All other values are RESERVED. Usage with CTA-861: When the Type X Timing Data Block is packed in a CTA Data Block, which separately indicates the length of the CTA Data Block payload, the "Number of Payload Bytes in Block" field shall not be present, and the size of the Type X Timing Data Block is reduced by 1 byte.
03h – 03h+M-1	(M*8-1):0	**First Type X Formula Timing Descriptor** M-byte descriptor. Priority 1.
03h+M – 03h+2M-1	(M*8-1):0	**Second Type X Formula Timing Descriptor** M-byte descriptor if present. Priority 2.
…		…
03h+(N-1)*M – 03h+(N*M)-1	(M*8-1):0	**Nth Type X Formula Timing Descriptor** M-byte descriptor. Priority N.

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Table 4-XY: Type X Formula-based Timing Descriptor Format

Byte #	Bit #	Description/Format	6-Byte	7-Byte
0	**Timing Options**		✔	✔
	2:0	**Timing Formula/Algorithm** 000b = *CVT* standard blanking. 001b = *CVT* reduced blanking version 1 (CVT RB v1). 010b = *CVT* reduced blanking version 2 (CVT RB v2). 011b = *CVT* reduced blanking version 3 (CVT RB v3). All other values are RESERVED.		
	3	**RESERVED** Cleared to 0.	✔	✔
	4	The definition of bit 4 meaning is dependent on Timing Formula/Algorithm field. If Timing Formula/Algorithm field is set to 010b (i.e. reduced blanking version 2) then the bit has following meaning: **CVT RB v2 ~~version 2~~: Video Refresh Rate Option** 0 = Refresh rate × (1000 / 1001) variant of timing is not supported. 1 = Refresh rate × (1000 / 1001) variant of timing is also supported, along with base Refresh rate. If Timing Formula/Algorithm field is set to 011b (i.e. reduced blanking version 3) then the bit has following meaning: **CVT RB v3 ~~version 3~~: Horizontal Blank Option** 0 = Horizontal Blank of 80 Pixels. 1 = Horizontal Blank of 160 Pixels. For all other values of Timing Formula/Algorithm field, the bit shall be cleared to 0b.	✔	✔
	6:5	**3D Stereo Support** 00b = This timing shall always be displayed in mono (no stereo). 01b = This timing shall always be displayed in stereo. 10b = This timing shall be displayed in mono or stereo, depending on a user action (such as wearing stereo glasses). 11b = RESERVED.	✔	✔
	7	**YCC 420 Support** 0 = No explicit YCC 420 support for current mode (other blocks may be utilized to indicate support) 1 = YCC 420 supported for current timing	✔	✔
2:1	**Horizontal Active Image Pixels** Pixel quantities range from 1 through 65,536.		✔	✔
1	7:0	**Horizontal Active Image Pixels Low Bits 7:0** Values range from 00h through FFh.	✔	✔
2	15:8	**Horizontal Active Image Pixels High Bits 15:8** Values range from 00h through FFh.	✔	✔

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4:3	**Vertical Active Image Lines** Line quantities range from 1 through 65,536.		✔	✔
3	7:0	**Vertical Active Image Lines Low Bits 7:0** Values range from 00h through FFh.	✔	✔
4	15:8	**Vertical Active Image Lines High Bits 15:8** Values range from 00h through FFh.	✔	✔
5	7:0	**Refresh Rate Low Bits 7:0** Least significant 8 bits of Refresh Rate field [bit0 …bit7]. Note: 6 Byte Type X Formula-based Timing Descriptor end at this offset and this limits the Refresh Rate field to 256Hz. As value of 0 correspond to 1Hz.	✔	✔
6	1:0	**Refresh Rate High Bits 9:8** Note: Defined only for 7 Byte Type X Formula-based Timing Descriptor. The value of the Refresh Rate field comprises of byte 5 bits [0:7] and byte 6 bits [0:1] to make up the 10 bits Refresh rate field. Refresh Rate has a range of 1 – 1024Hz, with value of 0 corresponds to 1 Hz an 3FFh corresponding to 1024 Hz.		✔
	7:2	**RESERVED** **Cleared to 0.**		✔

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6	1:0	**Refresh Rate High Bits 9:8** Note: Defined only for 7 Byte Type X Formula-based Timing Descriptor. The value of the Refresh Rate field comprises of byte 5 bits [0:7] and byte 6 bits [0:1] to make up the 10 bits Refresh rate field. Refresh Rate has a range of 1 – 1024Hz, with value of 0 corresponds to 1 Hz an 3FFh corresponding to 1024 Hz.		✔
	4:2	**Delta Horizontal Blank** Note: This field is only applicable for *CVT* reduced blanking timing version 3 formula reported in **Timing Formula/Algorithm** field. **The value range of the field is 0-7.** The field provides the delta horizontal blank pixels to be added on top of the value reported in **CVT RB v3 ~~version 3~~: Horizontal Blank Option;** notes the delta value is dependent on the field (i.e. 80 vs 160) to accommodate the flexibility range of 80-200 horizontal blank pixels. **If the CVT RB v3 ~~version 3~~: Horizontal Blank Option** is set to **"**Horizontal Blank of 80 Pixels" Horizontal Blank in Pixels = [Field Value] * 8 + 80 **If the CVT RB v3 ~~version 3~~: Horizontal Blank Option** is set to **"**Horizontal Blank of 160 Pixels" If Field Value <= 5 Horizontal Blank in Pixels = [Field Value] * 8 + 160 Else if 5 < Field Value <=7 Horizontal Blank in Pixels = [5-Field Value] * 8 + 160 Table below describes the various options that is possible for "Horizontal Blank" with combination of this field and Horizontal Blank Option in Timing Options Byte. **Field Value** [Byte 6 bits 4:2] / **Horizontal Blank in Pixels with Horizontal Blank Option = 0** / **Horizontal Blank in Pixels with Horizontal Blank Option = 1** 0 / 0*8 + 80 = 80 / 0*8 + 160 = 160 1 / 1*8 + 80 = 88 / 1*8 + 160 = 168 2 / 2*8 + 80 = 96 / 2*8 + 160 = 176 3 / 3*8 + 80 = 104 / 3*8 + 160 = 184 4 / 4*8 + 80 = 112 / 4*8 + 160 = 192 5 / 5*8 + 80 = 120 / 5*8 + 160 = 200 6 / 6*8 + 80 = 128 / (5-6)*8 + 160 = 152 7 / 7*8 + 80 = 136 / (5-7)*8 + 160 = 144		✔

	7:5	**Additional Vertical Blank Time in % of frame time for the defined Refresh Rate** Note: This field is only applicable for *CVT RB v3* ~~reduced blanking version 3~~ timing formula reported in **Timing Formula/Algorithm** field. If this field is not defined a Vertical Blank of 460us shall be used. **The value range of the field is 0-7.** The Additional Vertical Time = [Field Value] * 10000 / Refresh Rate provides the additional vertical blank time in microseconds to be added to Minimum Vertical Blank Time of 460us provided the overall blank time does not exceed 25% of the frame rate for the defined Refresh Rate. If the vertical blank time exceeds the 25%, then the default Minimum Vertical Blank Time of 460us shall be used. Blank Time to be used with CVT **RB v3** formula = 460us + Additional Vertical Blank Time, as long as the Total Blank Time is less than 25% of the Frame Time, otherwise the Blank Time is set to 460us. **The following tables shows few examples with 60Hz and 480Hz.**		✔

Field Value [Byte 6 bits 7:5]	Additional Vertical Time for **Refresh Rate = 60Hz** [Byte 5 bits 7:0, Byte 6 bits 1:0]	Additional Vertical Time for **Refresh Rate = 480 Hz** [Byte 5 bits7:0, Byte 6 bits 1:0]
0	0*10000/60 = 0	0*10000/480 = 0
1	1*10000/60 = 116.67	1*10000/480 = 20.83
2	2*10000/60 = 333.33	2*10000/480 = 41.67
3	3*10000/60 = 500.00	3*10000/480 = 62.50
4	4*10000/60 = 666.67	4*10000/480 = 83.33
5	5*10000/60 = 833.33	5*10000/480 = 104.17
6	6*10000/60 = 1000.00	6*10000/480 = 125.00
7	7*10000/60 = 1166.67	7*10000/480 = 145.83

With maximum allotment of 25% frame time for vertical blank; at 60Hz refresh rate all additional values when added to minimum blank of 460us will be well below the max allowed and hence all can be used. Allowing for a maximum vertical blank time of 1166.67 + 460 = 1626.67 us

However, at 480Hz refresh rate, 25% of the frame time is 520.83us. Which means that only field value of 0, 1 and 2 can be used since other values when added to minimum blank leads to a blank of more then 25% of frame time (i.e. blank time is greater than 520.83us). For values of 3-7, the Blank Time shall be clamped at 460us.

- End of Document -

VESA STANDARDS CHANGE REQUEST FORM

To be Filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	Mandatory DisplayID Data Blocks and Structure Coexistence
AFFECTED DOCUMENT:	DisplayID ver 2.0
REVISION CATEGORY:	Refer to VP235H Appendix A; will be subject to Task Group review
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Syed Athar Hussain, AMD

SCR REVISION HISTORY	
(DATE)	**(CHANGE)**
2/1/21	Initial Draft, rev1
3/31/21	Rev2, editorial changes to resolve GMR comments; added requirement that Sources be able to parse EDID extensions presented in any order
4/22/21	Rev3, modified conditions that mandate use of certain data blocks when DisplayID v2.0 is used as an EDID extension

(add more rows as needed)

To be Filled in by VESA Office:

VESA SCR NUMBER:	(To be assigned by VESA office)
SCR ENTRY DATE:	02/03/2021

To be Filled in by Task Group or VESA Office

SCR ADOPTED, REJECTED, or otherwise DISPOSITIONED for other action	SCR is (adopted) or (rejected) or (Dispositioned for other action) If rejected, explain reason for acceptation or rejection If dispositioned, explain action or plan for action (such as including in future draft specification revision, or re-visiting at future date, or other)
DATE SCR ADOPTED or REJECTED or DISPOSITIONED	ADOPTED 05/10/2021

Summary of the Proposed Change(s)

The SCR clarifies the mandatory DisplayID data blocks for DisplayID version 2.0 and beyond.

IPR (Intellectual Property Rights) declaration, if any

None

Benefits as a Result of the Changes

With the proposed SCR sink and source will have better interoperability as both source and sink expectation will be clarified with respect to mandatory and optional DisplayID data blocks. Furthermore, it will allow DisplayID CTS to be developed with those mandates so both sources and sinks will be tested against that.

Assessment of the Impact

SW/FW update on source and sinks side.

Analysis of the Device Hardware Implication

No hardware impact is expected.

Analysis of the Device Software Implications

SW and FW may need to be updated to support the mandatory data blocks.

Analysis of the Compliance Test & Interop Implications

DisplayID and EDID CTS would need to be updated to ensure data block mandates are being met on the sink side and source is able to parse the relevant blocks to enable specific features.

New Referenced Documents Resulting from Change

DisplayID ver 2.0

Attachments

None

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Proposed Document Change(s) or Addition(s)

Table 3-1 lists the data blocks that are defined as of *DisplayID v2.0*.

Table 3-1: Data Block Tag Allocation

Block Tag	Data Block Name[c]	Mandatory Block[a,b] (Native Structure)	Mandatory Block[a,b] (EDID Extension)	Defined in
1Fh – 00h	**RESERVED** RESERVED for legacy data blocks for DisplayID Structure *v1.x*. and shall **not** be used with new DisplayID Structure *v2.x.*			
20h	**Product Identification Data Block**	Yes, for Display Device	Yes, for Display Device if either Tiled Display Topology or Vendor-Specific Data Block (using other than VESA OUI) is present	Section 4.1
21h	**Display Parameters Data Block**	Yes, for Display Device	Yes, for Display Device	Section 4.2
22h	**Type VII Timing – Detailed Timing Data Block**	Yes, for Display Device; Must contain at least one timing	Yes, for Display Device; Must contain at least one timing	Section 4.3.1
23h	**Type VIII Timing – Enumerated Timing Code Data Block**	No	No	Section 4.3.2
24h	**Type IX Timing – Formula-based Timing Data Block**	No	No	Section 4.3.3
25h	**Dynamic Video Timing Range Limits Data Block**	No	No	Section 4.4
26h	**Display Interface Features Data Block**	Yes, for Display Device	Yes, for Display Device	Section 4.5
27h	**Stereo Display Interface Data Block**	No	No	Section 4.6
28h	**Tiled Display Topology Data Block**	Yes, for Display Device supporting Tiled Display	Yes, for Display Device supporting Tiled Display	Section 4.7
29h	**ContainerID Data Block**	Yes, for Multi-function Display Device	Yes, for Multi-function Display Device	Section 4.8

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2Ah	**Type X Timing – Formula-based Timing Data Block**	Yes, for Display Device supporting multiple CVT RB v3 timings.	No	
2Bh	**Adaptive-Sync Data Block**	Yes, for Display Device supporting Adaptive-Sync	Yes, for Display Device supporting Adaptive-Sync	
7Dh – 2Ch	**RESERVED for Additional VESA-defined Data Blocks**			
7Eh	**Vendor-specific Data Block**	No	No	Section 4.9
80h – 7Fh	**RESERVED**			
81h	**CTA Audio Data Block Encapsulated in DisplayID CTA Data Block** [c]	Yes, for Display Device with in-band Interface Audio Support beyond what can be exposed in DisplayID Data Blocks.	Yes (unless CTA extension is present), for Display Device with in-band Interface Audio Support beyond what can be exposed in DisplayID Data Blocks.	Section 4.10
81h	**CTA Speaker Allocation Block Encapsulated in DisplayID CTA Data Block** [c]	Yes, for Display Device with in-band Interface Audio Support beyond what can be exposed in DisplayID Data Blocks.	Yes (unless CTA extension is present), for Display Device with in-band Interface Audio Support beyond what can be exposed in DisplayID Data Blocks.	Section 4.10
81h	**CTA HDR Static Metadata Data Block Encapsulated in DisplayID CTA Data Block** [c]	Yes, for Display Device with HDR Support beyond what can be exposed in DisplayID Data Blocks.	Yes (unless CTA extension is present), for Display Device with HDR Support beyond what can be exposed in DisplayID Data Blocks.	Section 4.10
81h	**Other CTA Data Blocks Encapsulated in DisplayID CTA Data Block** [c]	No	No	Section 4.10

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FFh – 82h	**RESERVED** RESERVED for additional data blocks related to external standards organization(s).			

a. *"Yes, for Display Device" indicates that the Display Product Primary Use Case field of the DisplayID structure for the relevant product shall be populated with a value within the range of 2h through 8h.*
b. *Source that supports DisplayID version 2.0, shall be able to parse all Mandatory Data Blocks and enable relevant features based on source capability declaration form as part of Compliance Test Specification.*
c. *Source shall be able to parse encapsulated version of the CTA Data Blocks as well as CTA Data Blocks that exists in EDID CTA extension block.*

From DP2.0 Errata 7 SCR "Support DisplayID version 2 section as a base EDID extension."
A new paragraph will be added in Section 2.1 to describe how to set up DisplayID structure version 2 section as a base EDID extension. The proposed new paragraph is described below in red fonts.

2.1 DisplayID structure version 2.0 as a base EDID extension

To facilitate usage of DisplayID structure version 2.0 section(s) as base EDID extension(s), a new framework is defined.

EDIDs that contain extensions that use DisplayID structure version 2.0 shall include all mandatory blocks, as described in Table 3-1 and shall populate the listed fields as described below:

- EDID extension block tag shall be set to 70h denoting existence of a DisplayID section
- Following the block tag byte, a regular DisplayID structure version 2.0 fixed-length section shall be used to map to the EDID extension requirement of 128 bytes per extension.
- DisplayID Structure Version/Revision byte shall be set to 20h corresponding to DisplayID structure version 2 revision 0.
- First DisplayID structure version 2.0 section EDID extension shall populate the "Display Product Primary Use Case" field with a value from 1h-8h based on the intended primary use case of the sink. If the Primary Use Case is not known Sink shall set the value of 2h. Any subsequent DisplayID structure version 2.0 section EDID extension(s) shall set the Display Product Primary Use Case byte to 0h.
- Extension Count byte shall be set to 00h
- The "Bytes in Section" field shall be set to 79h (121). This corresponds to 5 additional bytes for the DisplayID section framework + EDID extension block tag for DisplayID and EDID extension block checksum for total of 128 bytes
- EDID extension block checksum shall be calculated using the DisplayID section checksum and extension block tag byte. Since DisplayID section checksum already ensures that all DisplayID

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section bytes adds up so that modulo 256 of the sums of all byte is 0h, only block tag byte needs to be considered for EDID extension block checksum. Consequently, EDID extension block checksum shall be set to 90h.

In order to facilitate compatibility with legacy Sources that may not have been updated to be able to parse DisplayID structure version 2.0, a Sink may expose separate DisplayID extensions with structure version 1.2 and structure version 2.0. If a Sink exposes extensions with both structure version 1.2 and structure version 2.0, then in addition to the requirements above for version 2.0 extensions, the following requirements shall be also be met:

- An EDID extension with DisplayID structure version 1.2 shall meet the requirements of the DisplayID v1.3 specification. In particular, note that all data blocks are optional for DisplayID EDID extensions with structure version 1.2 - which blocks to include is an implementation choice.
- EDID extensions with DisplayID structure version 2.0 shall be exposed at higher EDID block numbers than any optional DisplayID EDID extensions with structure version 1.2.
- EDID extensions (both DisplayID structure versions 1.2 and 2.0) shall be exposed at higher EDID block numbers than any optional CTA extensions.

Table X.X: Example of Required Ordering of EDID Extension Blocks

Segment Pointer / Address	EDID Block Number	Block Type
00h / 00h → 7Fh	0	Base EDID Structure
00h / 80h → FFh	1	Extension Block: CTA-861
01h / 00h → 7Fh	2	Extension Block: DisplayID Structure v1.2
01h / 80h → FFh	3	Extension Block: DisplayID Structure v2.0
02h / 00h → 7Fh	4	Extension Block: DisplayID Structure v2.0

A Source that supports parsing EDID extensions that use DisplayID structure version 2.0, shall parse EDID extensions regardless of where they appear within a valid EDID structure. In the case that similar capabilities are described in multiple EDID blocks, the Source shall use the information exposed in the DisplayID structure version 2.0 extension and ignore the duplicate or conflicting information exposed in other EDID blocks. Examples of similar capabilities are a Display Parameters Data Block that is present in both a DisplayID structure version 1.2 extension and a DisplayID structure version 2.0 extension, or chromaticity information that is present in both the EDID base block and a Display Parameters Data Block in a DisplayID structure version 2.0 extension.

- End of Document -

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VESA STANDARDS CHANGE REQUEST FORM

To be Filled in by Submitter (Refer to VESA Document VP235H, Section 5)

TITLE:	Created a new type X timing data block to support CVT ~~1.3~~ RB v3 and support refresh rate field up to 1024 Hz.
AFFECTED DOCUMENT:	DisplayID ver 2.0
REVISION CATEGORY:	Refer to VP235H Appendix A; will be subject to Task Group review
SUBMITTED TO:	DisplayID Task Group
SPONSOR:	Syed Athar Hussain, AMD

SCR REVISION HISTORY	
(DATE)	**(CHANGE)**
8/19/20	Initial Draft, rev1
8/24/20	Rev2, incorporate feedback from task group to allow 6-byte descriptor along with 7-byte descriptor.
9/8/20	Rev3, added provision to expose YCC420 support for each timing descriptor to accommodate CTA request.
11/06/20	Rev4, added provision to not include "Number of Payload Bytes in Block" byte when the block is packed in CTA block
2/01/21	Added new flexibility options for additional Horizontal Pixels and Vertical Blank Time to type X
2/10/21	Added tables and examples to clarify the field utilization for added flexibility.
3/31/21	Draft 7. Address GMR comments, changed references from CVT 1.3 to CVT RB v3 and fixed the example in page 4.
7/28/21	Draft 8. Remove the "Additional Vertical Blank Time in % of frame time for the defined Refresh Rate" as we support interoperable range up to 705us.

To be Filled in by VESA Office:

VESA SCR NUMBER:	(To be assigned by VESA office)
SCR ENTRY DATE:	03/03/2021

To be Filled in by Task Group or VESA Office

SCR ADOPTED, REJECTED, or otherwise DISPOSITIONED for other action	SCR is (adopted) or (rejected) or (Dispositioned for other action) If rejected, explain reason for acceptation or rejection If dispositioned, explain action or plan for action (such as including in future draft specification revision, or re-visiting at future date, or other)
DATE SCR ADOPTED or REJECTED or DISPOSITIONED	ADOPTED 08/16/2021

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Summary of the Proposed Change(s)

The SCR creates the type X data block to add CVT ~~1.3~~ RB v3 support and extend the Refresh Rate field to support up to 1024 Hz.

IPR (Intellectual Property Rights) declaration, if any

None

Benefits as a Result of the Changes

With the proposed updates, following benefit can be realized:

1. CVT ~~1.3~~ RB v3 reduced blanking version 3 timing can be exposed using type X Timing data block revision 0. Earlier type IX was limited to CVT 1.2 specification.
2. Type X data block version 0 will support up to 1024 Hz timing modes. Earlier type IX data block was limited to 256Hz timing.
3. Added Explicit YCC 420 per timing descriptor as per CTA request.
4. Added provision to not include the "Number of Payload Bytes in Block" byte when the Type X data block is packed in CTA block.

Assessment of the Impact

SW/FW update on source and sinks side.

Analysis of the Device Hardware Implication

No hardware impact is expected if CVT ~~1.3~~ RB v3 timing rules are supported.

No hardware impact if higher refresh rate timings are supported by sink and source. If not supported, then sink shall not expose the higher refresh rate timing.

Analysis of the Device Software Implications

SW and FW may need to be updated to support the new field associated with new data block revision described in the document.

Analysis of the Compliance Test & Interop Implications

Since CVT ~~1.3~~ RB v3 is being mandated for Adaptive-Sync. Adaptive-Sync related compliance test would need to be updated for source and sink testing.

New Referenced Documents Resulting from Change

DisplayID ver ~~2.1~~2.0

Attachments

None

Proposed Document Change(s) or Addition(s)

Background Information

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4.3.3 Type X Timing – Formula-based Data Block

The Type X Formula-based data block provides a means for a Sink device to expose timing support that is derived from an industry-standard formula. The revision 0 block supports *CVT* ~~*v1.3*~~ RB v3 timing with standard blanking, and reduced blanking version 1, version 2 and version 3. The 7-byte descriptor variant of the block also supports refresh rate field that extends to 1024 Hz. Note, the type X data block is a superset of old type IX data block and can accommodate all of the type IX capability exposure using a 6-byte descriptor variant, it is strongly recommended that new sink move to utilizing the Type X data block.

Sink may list its timing priority in the order of timing exposure as a hint, however, note source may not utilize sink exposed priority and instead implement a source implementation specific policy based on native resolution of the display and link capability.

Type X Timing is an **optional** data block.

Table 4-XX and Table 4-XY list the data block's structure and descriptor formats, respectively.

Table 4-XX: Type X Timing – Formula-based Data Block Format

Offset	Bit #	Description/Format/Priority
00h	7:0	**Type X Timing – Formula-based Timing Data Block** Value is 2Ah.
01h	**Block Revision and Other Data**	
	2:0	**Block Revision** Revision values range from 0 through 7. 000b = Revision 0. ~~001b = Revision 1.~~ All other values are RESERVED.
	3	Cleared to 0.
	6:4	**Number of Payload Bytes (M) in a Timing Descriptor** Where M (Bytes) = 6 (Initial descriptor size) + value of the field 000b = 6+0 bytes per descriptor (Defined as part of Type X, revision 0) 001b = 6+1 bytes per descriptor (Defined as part of Type X, revision 0) All other values are RESERVED.
	7	**RESERVED** Cleared to 0.

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02h	7:0	**Number of Payload Bytes in Block** Number of payload bytes within the block is based on # of descriptor (N) x Size of each descriptor (M) bytes. Note: value of M is based on "**Number of Payload Bytes (M) in a Timing Descriptor".** Source implementation shall accommodate varying field for future extensibility. All other values are RESERVED. Usage with CTA-861: When the Type X Timing Data Block is packed in a CTA Data Block, which separately indicates the length of the CTA Data Block payload, the "Number of Payload Bytes in Block" field shall not be present, and the size of the Type X Timing Data Block is reduced by 1 byte.
03h – 03h+M-1	(M*8-1):0	**First Type X Formula Timing Descriptor** M-byte descriptor. Priority 1.
03h+M – 03h+2M-1	(M*8-1):0	**Second Type X Formula Timing Descriptor** M-byte descriptor if present. Priority 2.
…		…
03h+(N-1)*M – 03h+(N*M)-1	(M*8-1):0	**Nth Type X Formula Timing Descriptor** M-byte descriptor. Priority N.

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Table 4-XY: Type X Formula-based Timing Descriptor Format

Byte #	Bit #	Description/Format	6-Byte	7-Byte
0	**Timing Options**		✔	✔
	2:0	**Timing Formula/Algorithm** 000b = *CVT* standard blanking. 001b = *CVT* reduced blanking version 1 (CVT RB v1). 010b = *CVT* reduced blanking version 2 (CVT RB v2). 011b = *CVT* reduced blanking version 3 (CVT RB v3). All other values are RESERVED.		
	3	**RESERVED** Cleared to 0.	✔	✔
	4	The definition of bit 4 meaning is dependent on Timing Formula/Algorithm field. If Timing Formula/Algorithm field is set to 010b (i.e. reduced blanking version 2) then the bit has following meaning: **CVT RB v2 ~~version 2~~: Video Refresh Rate Option** 0 = Refresh rate × (1000 / 1001) variant of timing is not supported. 1 = Refresh rate × (1000 / 1001) variant of timing is also supported, along with base Refresh rate. If Timing Formula/Algorithm field is set to 011b (i.e. reduced blanking version 3) then the bit has following meaning: **CVT RB v3 ~~version 3~~: Horizontal Blank Option** 0 = Horizontal Blank of 80 Pixels. 1 = Horizontal Blank of 160 Pixels. For all other values of Timing Formula/Algorithm field, the bit shall be cleared to 0b.	✔	✔
	6:5	**3D Stereo Support** 00b = This timing shall always be displayed in mono (no stereo). 01b = This timing shall always be displayed in stereo. 10b = This timing shall be displayed in mono or stereo, depending on a user action (such as wearing stereo glasses). 11b = RESERVED.	✔	✔
	7	**YCC 420 Support** 0 = No explicit YCC 420 support for current mode (other blocks may be utilized to indicate support) 1 = YCC 420 supported for current timing	✔	✔
2:1	**Horizontal Active Image Pixels** Pixel quantities range from 1 through 65,536.		✔	✔
1	7:0	**Horizontal Active Image Pixels Low Bits 7:0** Values range from 00h through FFh.	✔	✔
2	15:8	**Horizontal Active Image Pixels High Bits 15:8** Values range from 00h through FFh.	✔	✔

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4:3	**Vertical Active Image Lines** Line quantities range from 1 through 65,536.		✔	✔
3	7:0	**Vertical Active Image Lines Low Bits 7:0** Values range from 00h through FFh.	✔	✔
4	15:8	**Vertical Active Image Lines High Bits 15:8** Values range from 00h through FFh.	✔	✔
5	7:0	**Refresh Rate Low Bits 7:0** Least significant 8 bits of Refresh Rate field [bit0 …bit7]. Note: 6 Byte Type X Formula-based Timing Descriptor end at this offset and this limits the Refresh Rate field to 256Hz. As value of 0 correspond to 1Hz.	✔	✔
6	1:0	**Refresh Rate High Bits 9:8** Note: Defined only for 7 Byte Type X Formula-based Timing Descriptor. The value of the Refresh Rate field comprises of byte 5 bits [0:7] and byte 6 bits [0:1] to make up the 10 bits Refresh rate field. Refresh Rate has a range of 1 – 1024Hz, with value of 0 corresponds to 1 Hz an 3FFh corresponding to 1024 Hz.		✔
	7:2	**RESERVED** **Cleared to 0.**		✔

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<table>
<tr><td>6</td><td>1:0</td><td>Refresh Rate High Bits 9:8

Note: Defined only for 7 Byte Type X Formula-based Timing Descriptor.

The value of the Refresh Rate field comprises of byte 5 bits [0:7] and byte 6 bits [0:1] to make up the 10 bits Refresh rate field. Refresh Rate has a range of 1 – 1024Hz, with value of 0 corresponds to 1 Hz an 3FFh corresponding to 1024 Hz.</td><td></td><td>✔</td></tr>
<tr><td></td><td>4:2</td><td>Delta Horizontal Blank

Note: This field is only applicable for CVT reduced blanking timing version 3 formula reported in Timing Formula/Algorithm field.

The value range of the field is 0-7.

The field provides the delta horizontal blank pixels to be added on top of the value reported in CVT RB v3 ~~version 3~~: Horizontal Blank Option; notes the delta value is dependent on the field (i.e. 80 vs 160) to accommodate the flexibility range of 80-200 horizontal blank pixels.

If the CVT RB v3 ~~version 3~~: Horizontal Blank Option is set to “Horizontal Blank of 80 Pixels”

Horizontal Blank in Pixels = [Field Value] * 8 + 80

If the CVT RB v3 ~~version 3~~: Horizontal Blank Option is set to “Horizontal Blank of 160 Pixels”

If Field Value <= 5

Horizontal Blank in Pixels = [Field Value] * 8 + 160

Else if 5 < Field Value <=7

Horizontal Blank in Pixels = [5-Field Value] * 8 + 160

Table below describes the various options that is possible for “Horizontal Blank” with combination of this field and Horizontal Blank Option in Timing Options Byte.
<table>
<tr><th>Field Value
[Byte 6 bits 4:2]</th><th>Horizontal Blank in Pixels with Horizontal Blank Option = 0</th><th>Horizontal Blank in Pixels with Horizontal Blank Option = 1</th></tr>
<tr><td>0</td><td>0*8 + 80 = 80</td><td>0*8 + 160 = 160</td></tr>
<tr><td>1</td><td>1*8 + 80 = 88</td><td>1*8 + 160 = 168</td></tr>
<tr><td>2</td><td>2*8 + 80 = 96</td><td>2*8 + 160 = 176</td></tr>
<tr><td>3</td><td>3*8 + 80 = 104</td><td>3*8 + 160 = 184</td></tr>
<tr><td>4</td><td>4*8 + 80 = 112</td><td>4*8 + 160 = 192</td></tr>
<tr><td>5</td><td>5*8 + 80 = 120</td><td>5*8 + 160 = 200</td></tr>
<tr><td>6</td><td>6*8 + 80 = 128</td><td>(5-6)*8 + 160 = 152</td></tr>
<tr><td>7</td><td>7*8 + 80 = 136</td><td>(5-7)*8 + 160 = 144</td></tr>
</table></td><td></td><td>✔</td></tr>
</table>

	7:5	**RESERVED** Cleared to 0. Note in future the field will be utilized for defining Additional Vertical Blank requirement. ~~**Additional Vertical Blank Time in % of frame time for the defined Refresh Rate**~~ ~~Note: This field is only applicable for *CVT RB v3* reduced blanking version 3 timing formula reported in **Timing Formula/Algorithm** field. If this field is not defined a Vertical Blank of 460us shall be used.~~ ~~**The value range of the field is 0-7.**~~ ~~The Additional Vertical Time = [Field Value] * 10000 / Refresh Rate provides the additional vertical blank time in microseconds to be added to Minimum Vertical Blank Time of 460us provided the overall blank time does not exceed 25% of the frame rate for the defined Refresh Rate. If the vertical blank time exceeds the 25%, then the default Minimum Vertical Blank Time of 460us shall be used.~~ ~~Blank Time to be used with CVT **RB v3** formula = 460us + Additional Vertical Blank Time, as long as the Total Blank Time is less than 25% of the Frame Time, otherwise the Blank Time is set to 460us.~~ ~~**The following tables shows few examples with 60Hz and 480Hz.**~~		✓

~~**Field Value**~~ ~~[Byte 6 bits 7:5]~~	~~Additional Vertical Time for **Refresh Rate = 60Hz** [Byte 5 bits 7:0, Byte 6 bits 1:0]~~	~~Additional Vertical Time for **Refresh Rate = 480 Hz** [Byte 5 bits7:0, Byte 6 bits 1:0]~~
~~0~~	~~0*10000/60 = 0~~	~~0*10000/480 = 0~~
~~1~~	~~1*10000/60 = 116.67~~	~~1*10000/480 = 20.83~~
~~2~~	~~2*10000/60 = 333.33~~	~~2*10000/480 = 41.67~~
~~3~~	~~3*10000/60 = 500.00~~	~~3*10000/480 = 62.50~~
~~4~~	~~4*10000/60 = 666.67~~	~~4*10000/480 = 83.33~~
~~5~~	~~5*10000/60 = 833.33~~	~~5*10000/480 = 104.17~~
~~6~~	~~6*10000/60 = 1000.00~~	~~6*10000/480 = 125.00~~
~~7~~	~~7*10000/60 = 1166.67~~	~~7*10000/480 = 145.83~~

~~With maximum allotment of 25% frame time for vertical blank; at 60Hz refresh rate all additional values when added to minimum blank of 460us will be well below the max allowed and hence all can be used. Allowing for a maximum vertical blank time of 1166.67 + 460 = 1626.67 us~~

~~However, at 480Hz refresh rate, 25% of the frame time is 520.83us. Which means that only field value of 0, 1 and 2 can be used since other values when added to minimum blank leads to a blank of more then 25% of frame time (i.e. blank time is greater than 520.83us). For values of 3-7, the Blank Time shall be clamped at 460us.~~

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